JackBraun01 / helpdesk-ticketing-lab

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Self-hosted GLPI ticketing system integrated with Active Directory via encrypted LDAPS, extending the active-directory-lab project

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JackBraun01 Fix screenshot filenames and image paths db08a10 · 4 hours ago

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Troubleshooting	Fix screenshot filenames and image paths	4 hours ago
.gitignore	Initial commit: TICKET01 helpdesk lab with L...	4 hours ago
README.md	Initial commit: TICKET01 helpdesk lab with L...	4 hours ago
rename_script.ps1	Initial commit: TICKET01 helpdesk lab with L...	4 hours ago

Helpdesk Ticketing Lab: TICKET01

A self-hosted GLPI ticketing system built in VirtualBox and integrated with the existing `lab.local` Active Directory environment, simulating a small company's IT service desk — deployment, LDAP integration (upgraded to encrypted LDAPS with a real internal CA), and a realistic ticket set drawn from genuine troubleshooting encountered while building the environment.

Status: complete, including the LDAPS stretch goal — see Section 10 for the full write-up and Issues 7-11 in the Troubleshooting Log.

This project extends active-directory-lab — TICKET01 joins DC01 and CLIENT01 on the same isolated network, authenticating against the same domain.

Project Goal

To add a service-desk layer on top of an existing enterprise IT environment: deploy an open-source ITSM tool, integrate it with Active Director for real user authentication, and populate it with a realistic ticket history — including tickets drawn directly from genuine issues hit while building the infrastructure itself.

Environment Overview

Component	Details
Hypervisor	Oracle VirtualBox 7.2.16
Ticketing Server	`TICKET01` , Ubuntu Server 26.04.1 LTS (Resolute Raccoon), minimized install
Ticketing Software	GLPI (official `glpi/glpi` Docker image) + MySQL, via Docker Compose
Domain	Joins existing `lab.local` forest (see active-directory-lab)
Networking	Dual-homed: NAT (management/internet access) + isolated `LabNetwork` (same as DC01/CLIENT01)
TICKET01 LabNetwork IP	`192.168.1.30` (static, via netplan)
DNS	Points to DC01 (`192.168.1.10`)
Remote access	SSH from Windows PowerShell, via NAT port-forward (host `2222` → guest `22`)
GLPI web access	Browser via NAT port-forward (host `8080` → guest `80`)

Component	Details
LDAP Source	`lab.local` , authenticated against DC01

TICKET01 sits on the same isolated internal network as the existing domain controller and client, reflecting how a real internal service-desk server would sit inside a company's LAN rather than being a bolt-on SaaS tool. It is **dual-homed** — one network adapter for host managemer (NAT), a second dedicated purely to talking to the domain (LabNetwork) — the same pattern real servers use to separate management traffic from production traffic.

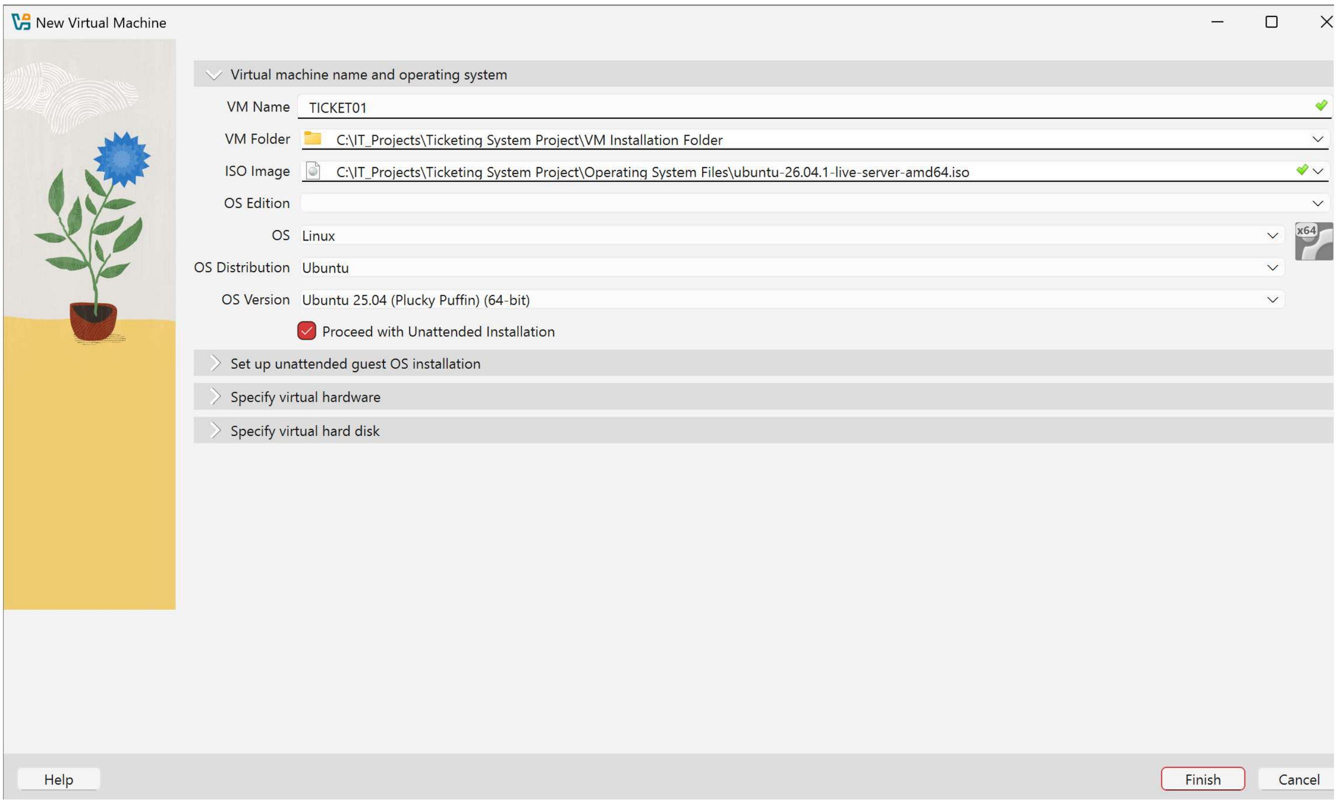
Design Rationale

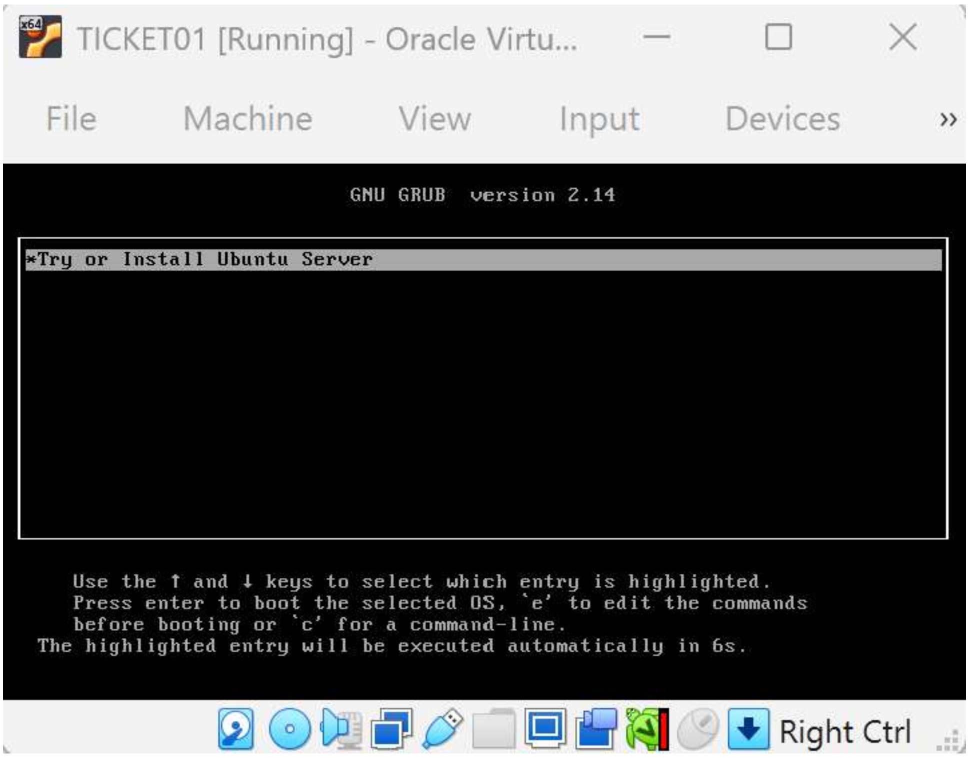
GLPI was chosen over other open-source options (osTicket, Zammad) specifically because it combines ticket management with native LDAP/Active Directory authentication — allowing genuine integration with the existing domain rather than a standalone tool with its own separate user base. This mirrors how commercial ITSM platforms commonly referenced in UK IT support job listings (ServiceNow, Freshservice integrate with corporate directory services.

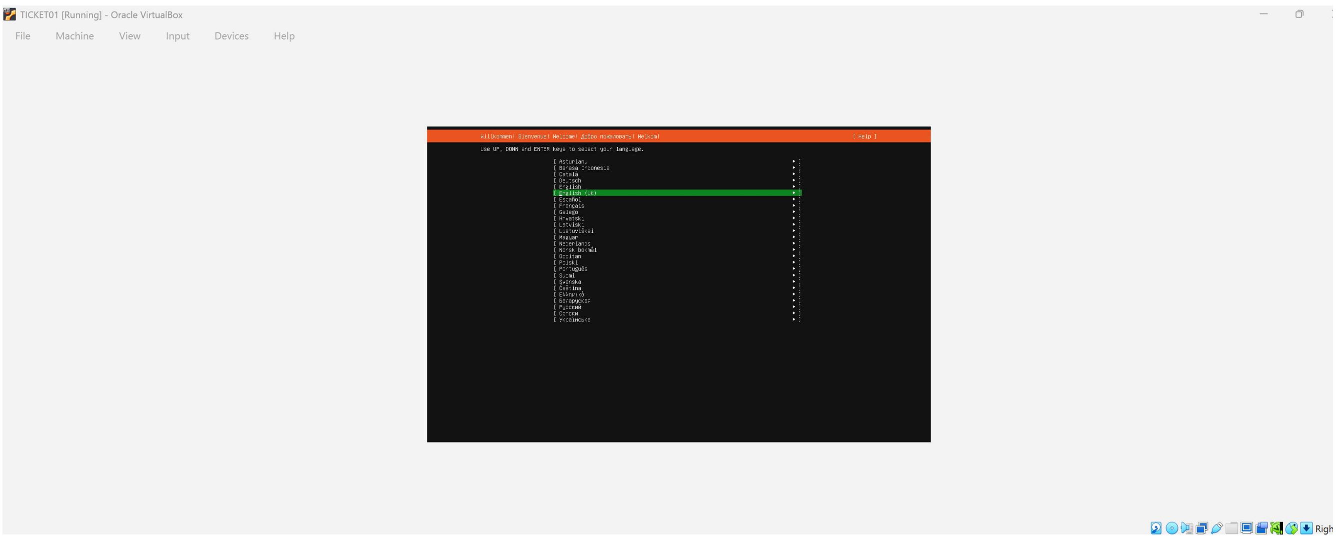
What Was Built

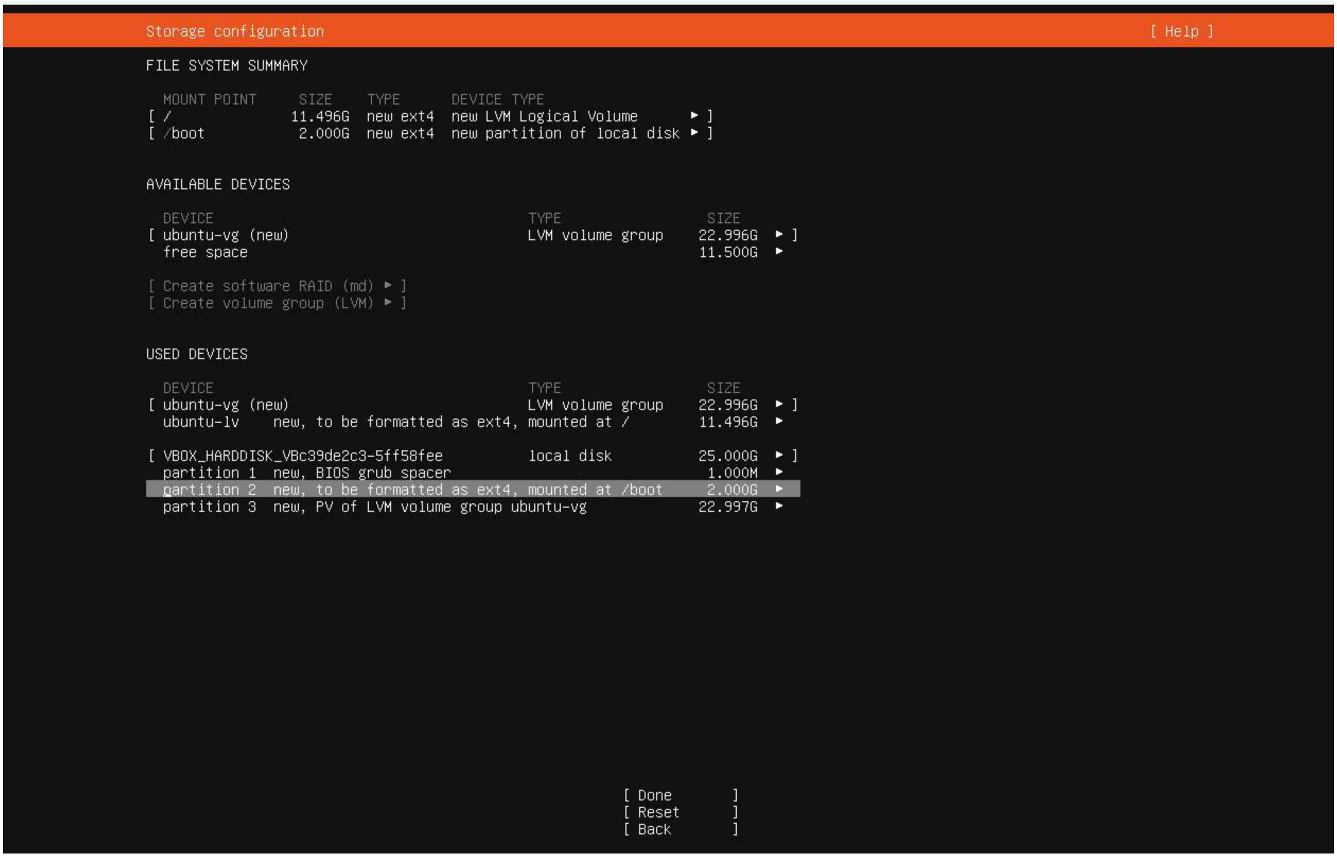
1. TICKET01 VM provisioning

Ubuntu Server 26.04.1 LTS installed via VirtualBox unattended installation (minimized install), initially on NAT networking to allow package/image downloads.

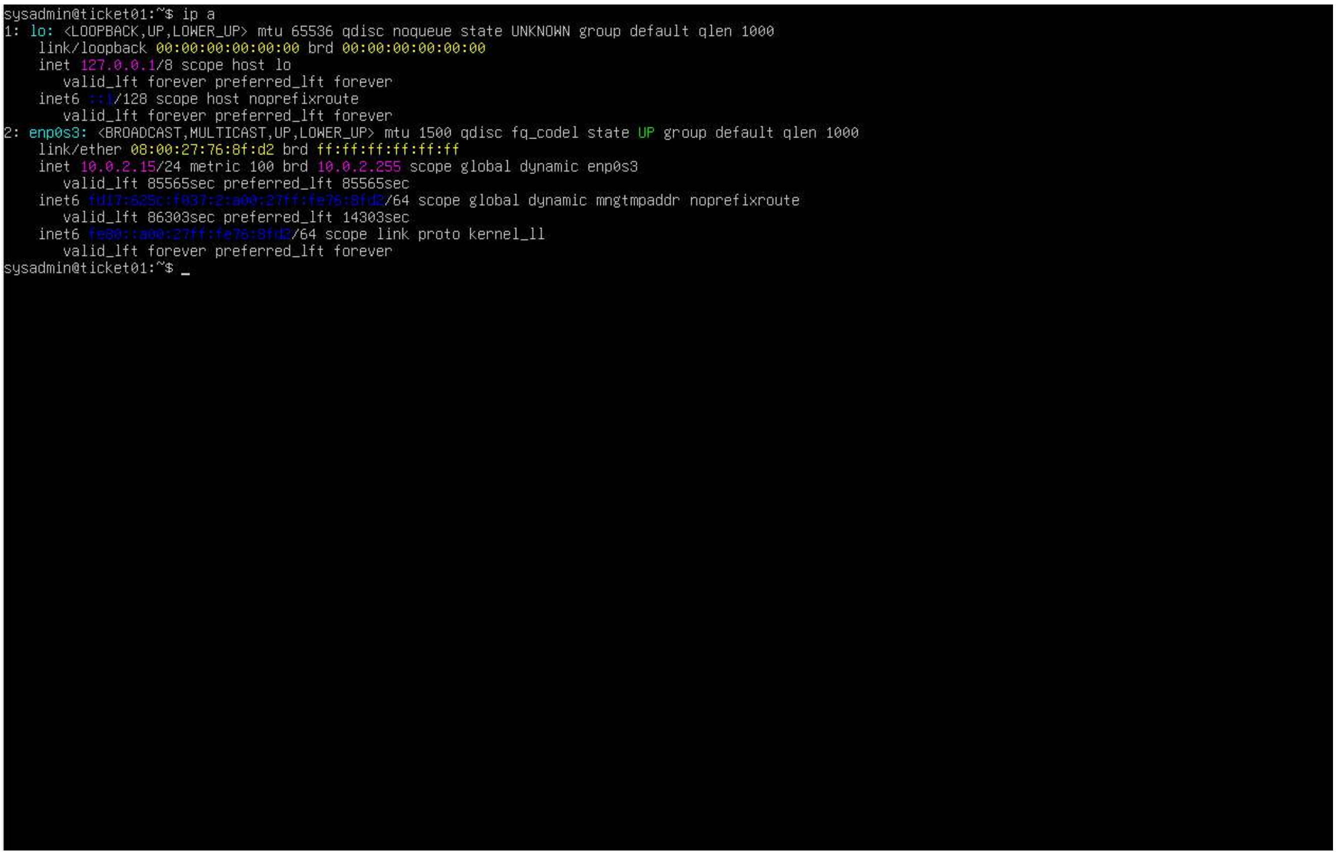

New Virtual Machine
Virtual machine name and operating system
VM Name TICKET01
VM Folder C:\IT_Projects\Ticketing System Project\VM Installation Folder
ISO Image C:\IT_Projects\Ticketing System Project\Operating System Files\ubuntu-26.04.1-live-server-amd64.iso
OS Edition
OS Linux
OS Distribution Ubuntu
OS Version Ubuntu 25.04 (Plucky Puffin) (64-bit)
Proceed with Unattended Installation
Set up unattended guest OS installation
Specify virtual hardware
Specify virtual hard disk
Help
Finish
Cancel


TICKET01 [Running] - Oracle Virtu...
File Machine View Input Devices
GNU GRUB version 2.14
*Try or Install Ubuntu Server
Use the ↑ and ↓ keys to select which entry is highlighted.
Press enter to boot the selected OS, `e' to edit the commands
before booting or `c' for a command-line.
The highlighted entry will be executed automatically in 6s.
Right Ctrl


TICKET01 [Running] - Oracle VirtualBox
File Machine View Input Devices Help

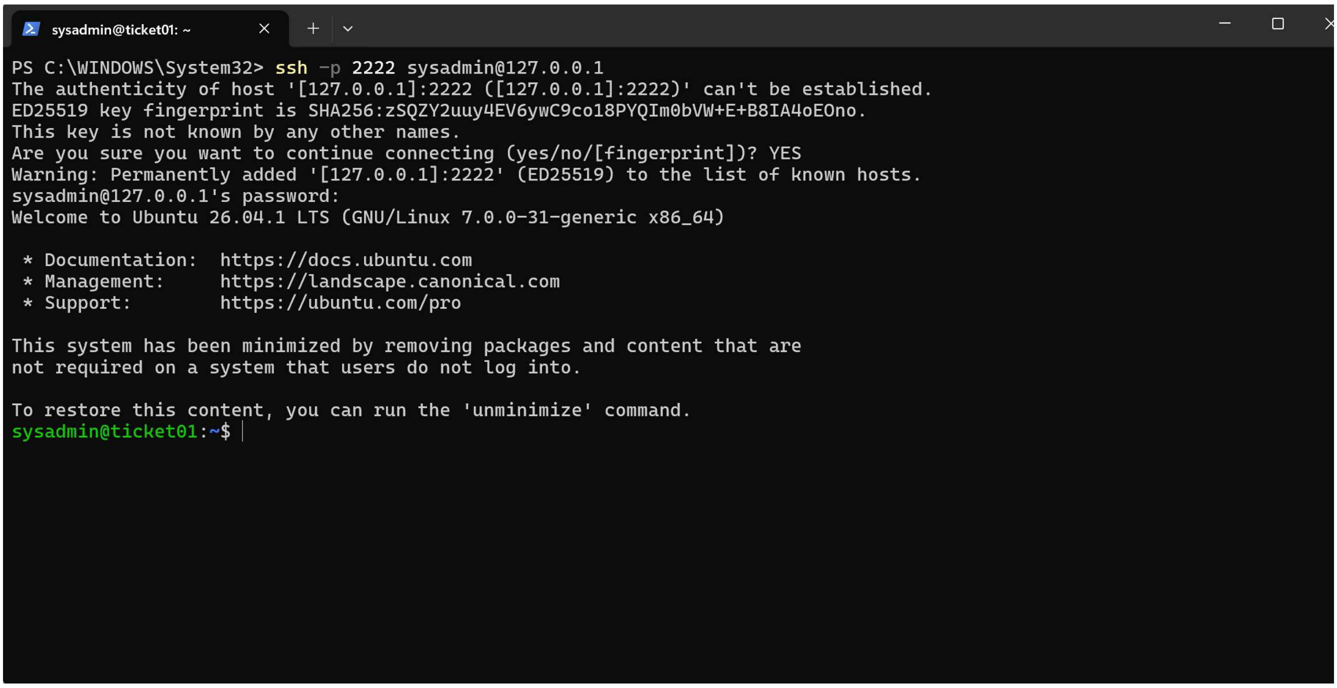

Storage configuration
[Help]
FILE SYSTEM SUMMARY
MOUNT POINT SIZE TYPE DEVICE TYPE
[/ 11.496G new ext4 new LVM Logical Volume ▸]
[/boot 2.000G new ext4 new partition of local disk ▸]
AVAILABLE DEVICES
DEVICE TYPE SIZE
[ubuntu-vg (new) LVM volume group 22.996G ▸]
free space 11.500G ▸
[Create software RAID (md) ▸]
[Create volume group (LVM) ▸]
USED DEVICES
DEVICE TYPE SIZE
[ubuntu-vg (new) LVM volume group 22.996G ▸]
ubuntu-lv new, to be formatted as ext4, mounted at / 11.496G ▸
[VBOX_HARDDISK_VBc39de2c3-5ff58fee local disk 25.000G ▸]
partition 1 new, BIOS grub spacer 1.000M ▸
partition 2 new, to be formatted as ext4, mounted at /boot 2.000G ▸
partition 3 new, PV of LVM volume group ubuntu-vg 22.997G ▸
[Done]
[Reset]
[Back]


sysadmin@ticket01:~$ _



2. Remote administration via SSH

OpenSSH server installed during setup; administered remotely from a Windows PowerShell client via a NAT port-forwarding rule (host `2222` - guest `22`), rather than the local VirtualBox console — reflecting standard real-world headless server management.



3. Docker & GLPI deployment

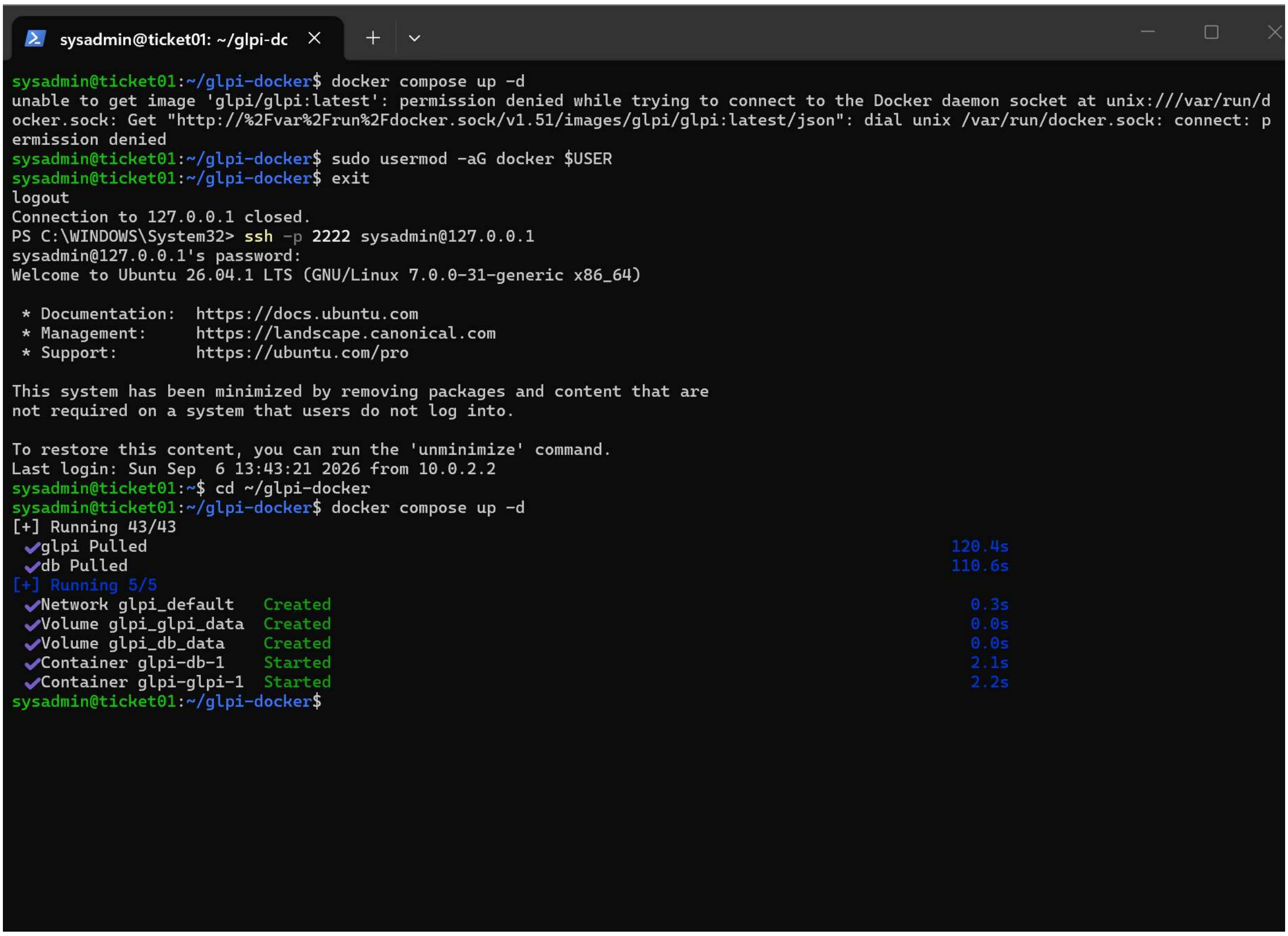
Docker Engine and Docker Compose installed via Ubuntu's official package repository. GLPI deployed using the official `glpi/glpi` Docker image maintained by the GLPI project, paired with a MySQL database container, defined via the project's official `docker-compose.example.yml` and `.env.example` templates.

sysadmin@ticket01: ~

```
er.socket'.
Setting up dnsmasq-base (2.92-1ubuntu0.4) ...
Setting up ubuntu-fan (0.12.17) ...
Created symlink '/etc/systemd/system/multi-user.target.wants/ubuntu-fan.service' → '/usr/lib/syst
tem/ubuntu-fan.service'.
Processing triggers for dbus (1.16.2-2ubuntu4) ...
Processing triggers for libc-bin (2.43-2ubuntu2.3) ...
Scanning processes...
Scanning candidates...
Scanning linux images...

Running kernel seems to be up-to-date.

Restarting services...

Service restarts being deferred:
 /etc/needrestart/restart.d/dbus.service
 systemctl restart unattended-upgrades.service

No containers need to be restarted.

User sessions running outdated binaries:
 sysadmin @ session #1: login[925]
 sysadmin @ session #3: sshd-session[2269]
 sysadmin @ user manager: (sd-pam)[1909]

No VM guests are running outdated hypervisor (qemu) binaries on this host.
sysadmin@ticket01:~$ sudo systemctl enable --now docker
sysadmin@ticket01:~$ docker --version
docker compose version
Docker version 29.1.3, build 29.1.3-0ubuntu4.1
Docker Compose version 2.40.3+ds1-0ubuntu1
sysadmin@ticket01:~$
```

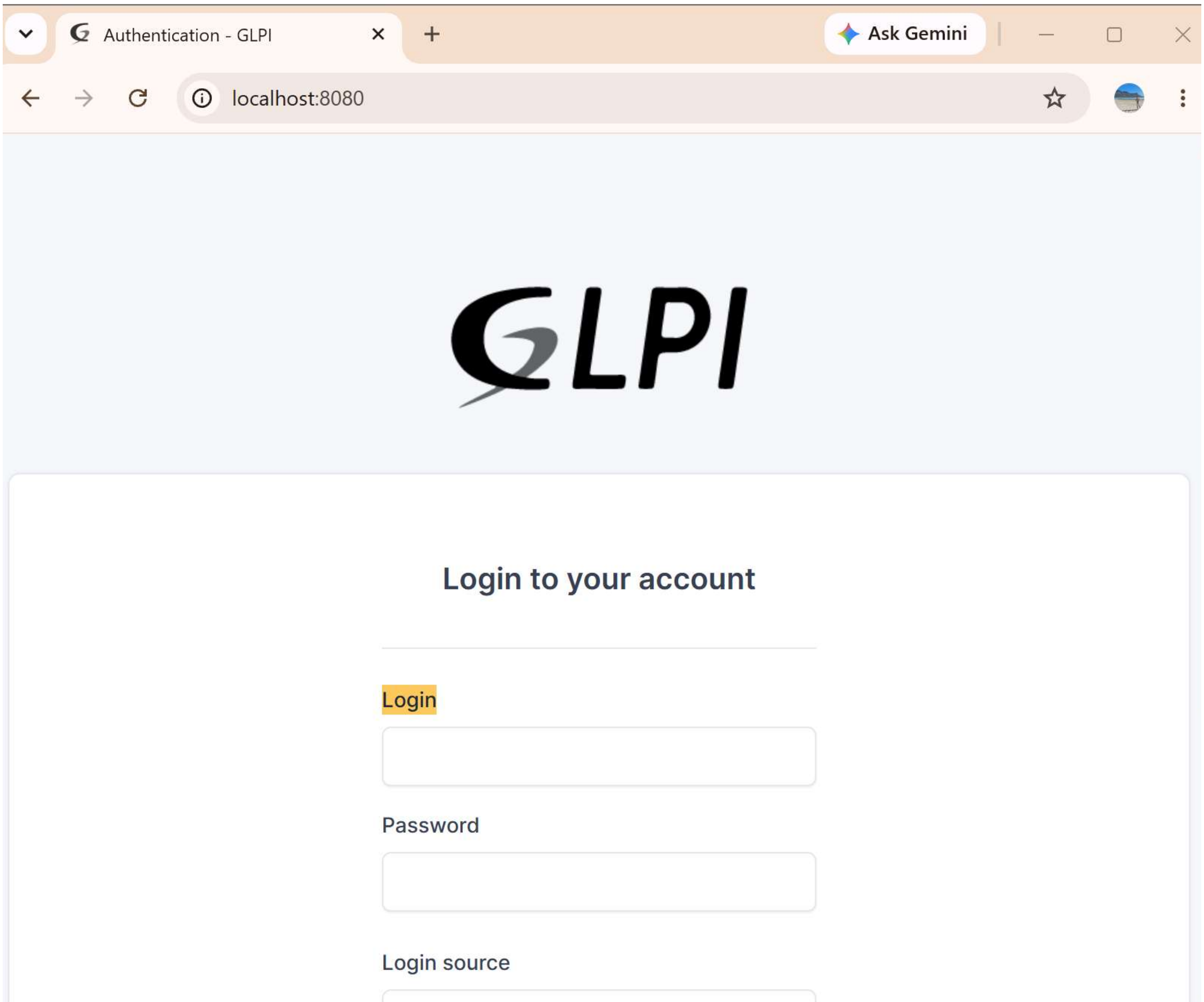
sysadmin@ticket01: ~/glpi-dc

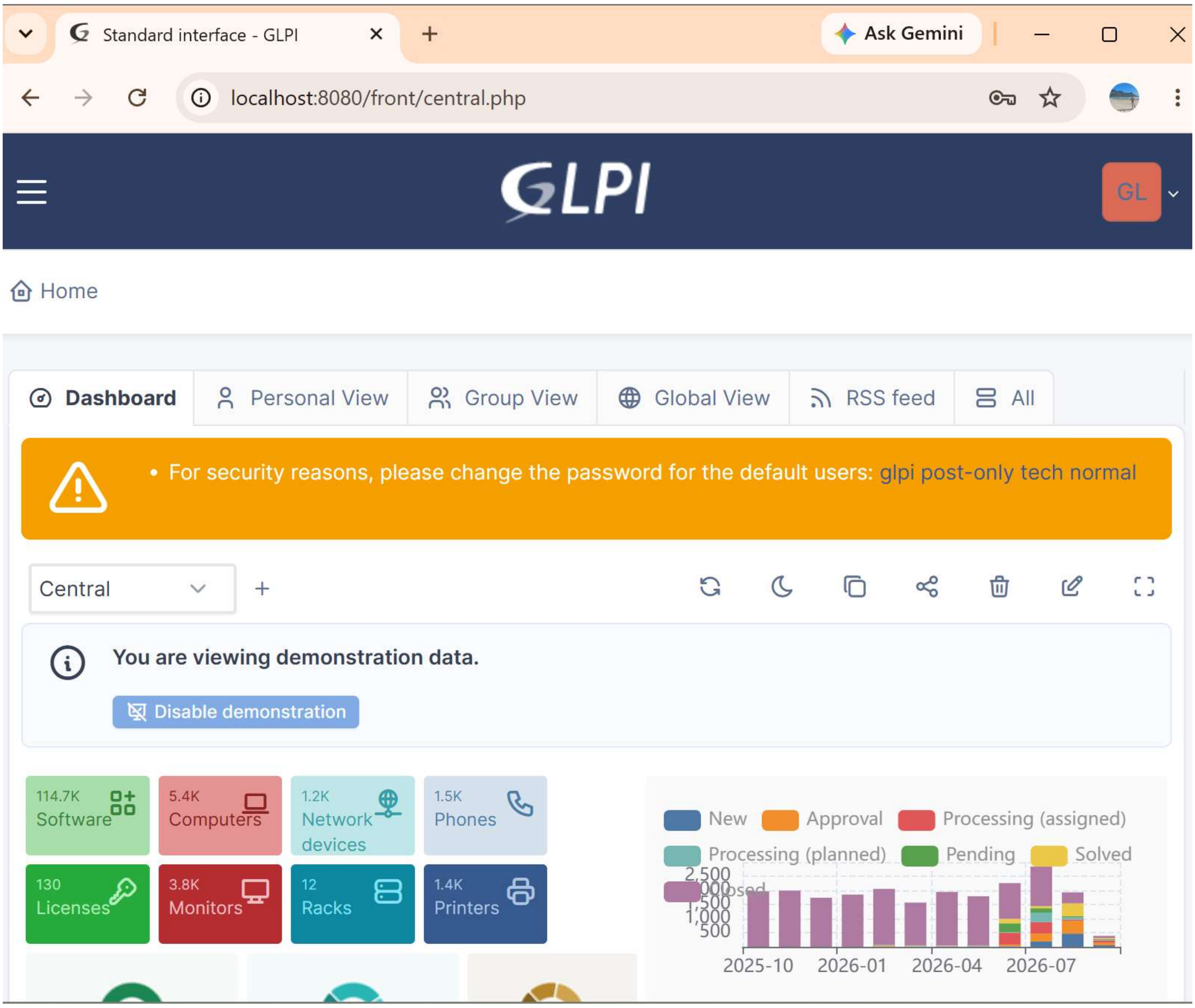
```
Get:1 http://gb.archive.ubuntu.com/ubuntu resolute-updates/main amd64 nano amd64 8.7.1-1ubuntu0.1
B]
Fetched 289 kB in 0s (1307 kB/s)
debconf: unable to initialize frontend: Dialog
debconf: (No usable dialog-like program is installed, so the dialog based frontend cannot be used
sr/share/perl5/Debconf/FrontEnd/Dialog.pm line 79, <STDIN> line 1.)
debconf: falling back to frontend: Readline
Selecting previously unselected package nano.
(Reading database ... 128535 files and directories currently installed.)
Preparing to unpack .../nano_8.7.1-1ubuntu0.1_amd64.deb ...
Unpacking nano (8.7.1-1ubuntu0.1) ...
Setting up nano (8.7.1-1ubuntu0.1) ...
update-alternatives: using /bin/nano to provide /usr/bin/editor (editor) in auto mode
update-alternatives: warning: skip creation of /usr/share/man/man1/editor.1.gz because associated
usr/share/man/man1/nano.1.gz (of link group editor) does not exist
update-alternatives: using /bin/nano to provide /usr/bin/pico (pico) in auto mode
update-alternatives: warning: skip creation of /usr/share/man/man1/pico.1.gz because associated f
r/share/man/man1/nano.1.gz (of link group pico) does not exist
Scanning processes...
Scanning linux images...

Running kernel seems to be up-to-date.

No services need to be restarted.

No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
sysadmin@ticket01:~/glpi-docker$ nano .env
sysadmin@ticket01:~/glpi-docker$
```



4. First GLPI access

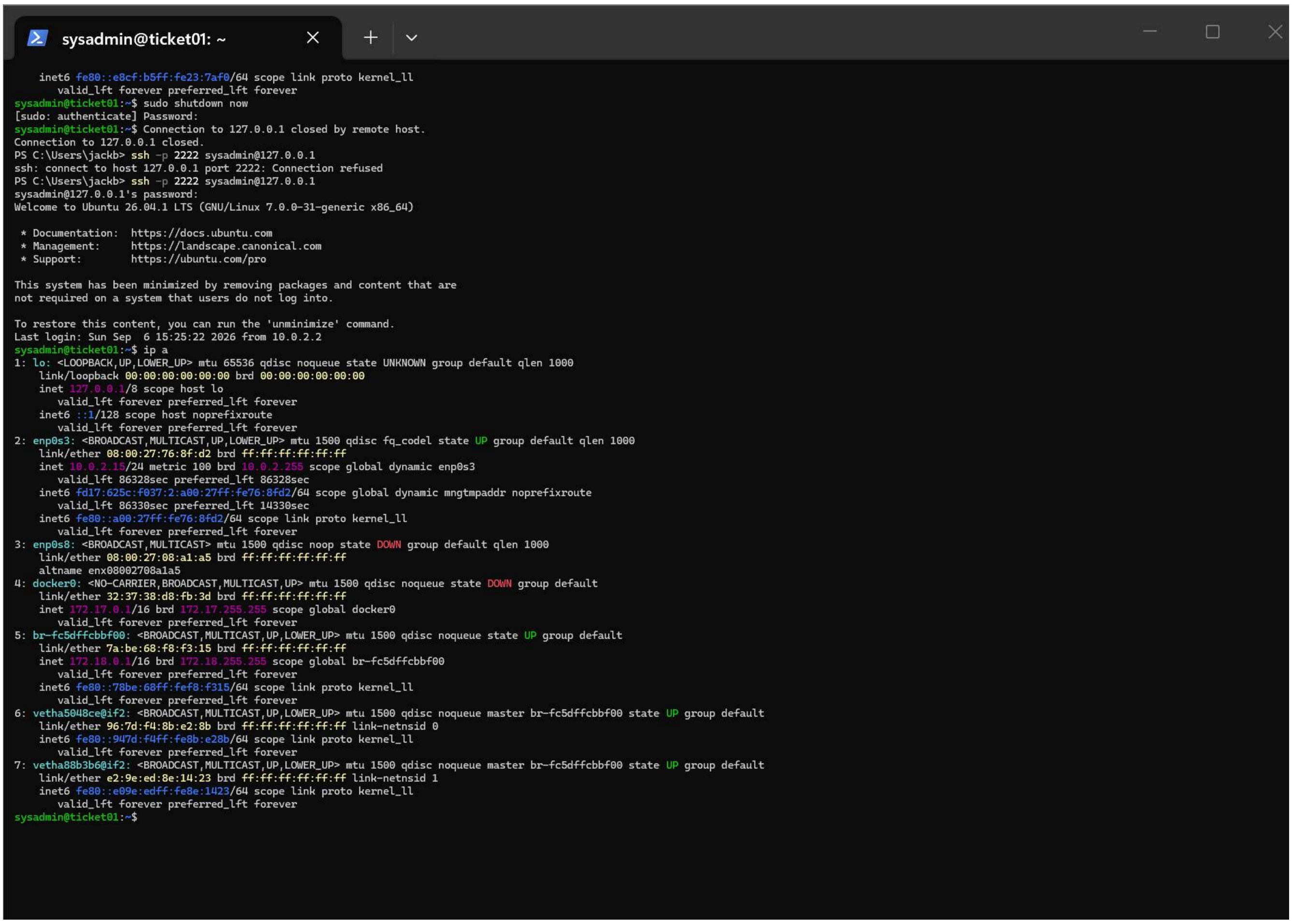
A second NAT port-forwarding rule (host `8080` → guest `80`) exposed GLPI's web interface to the host browser. GLPI auto-installed its own database schema on first run (detected via the `.env` database variables), skipping straight to a working login screen.

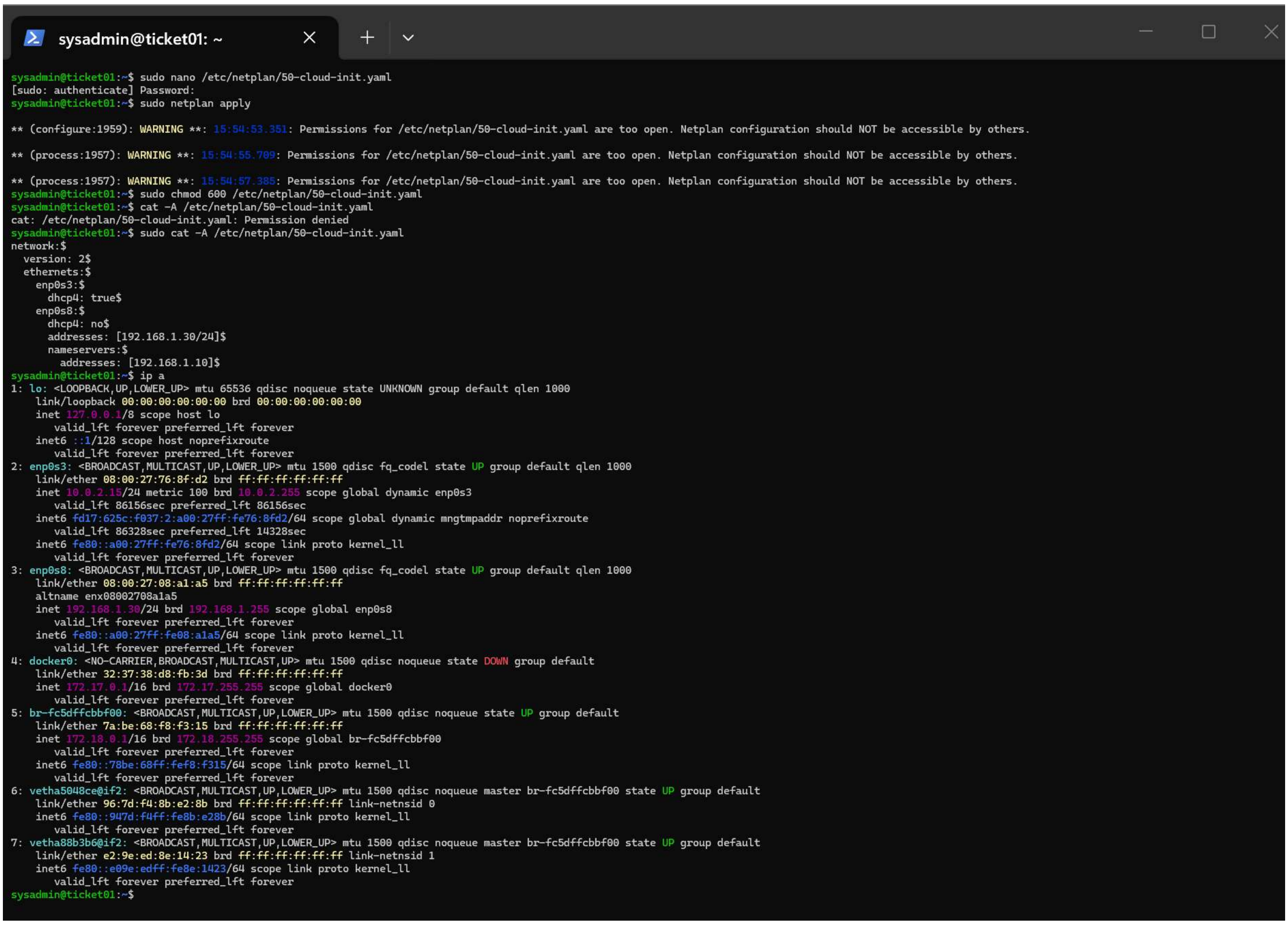

Authentication - GLPI
Ask Gemini
localhost:8080
Login to your account
Login
Password
Login source

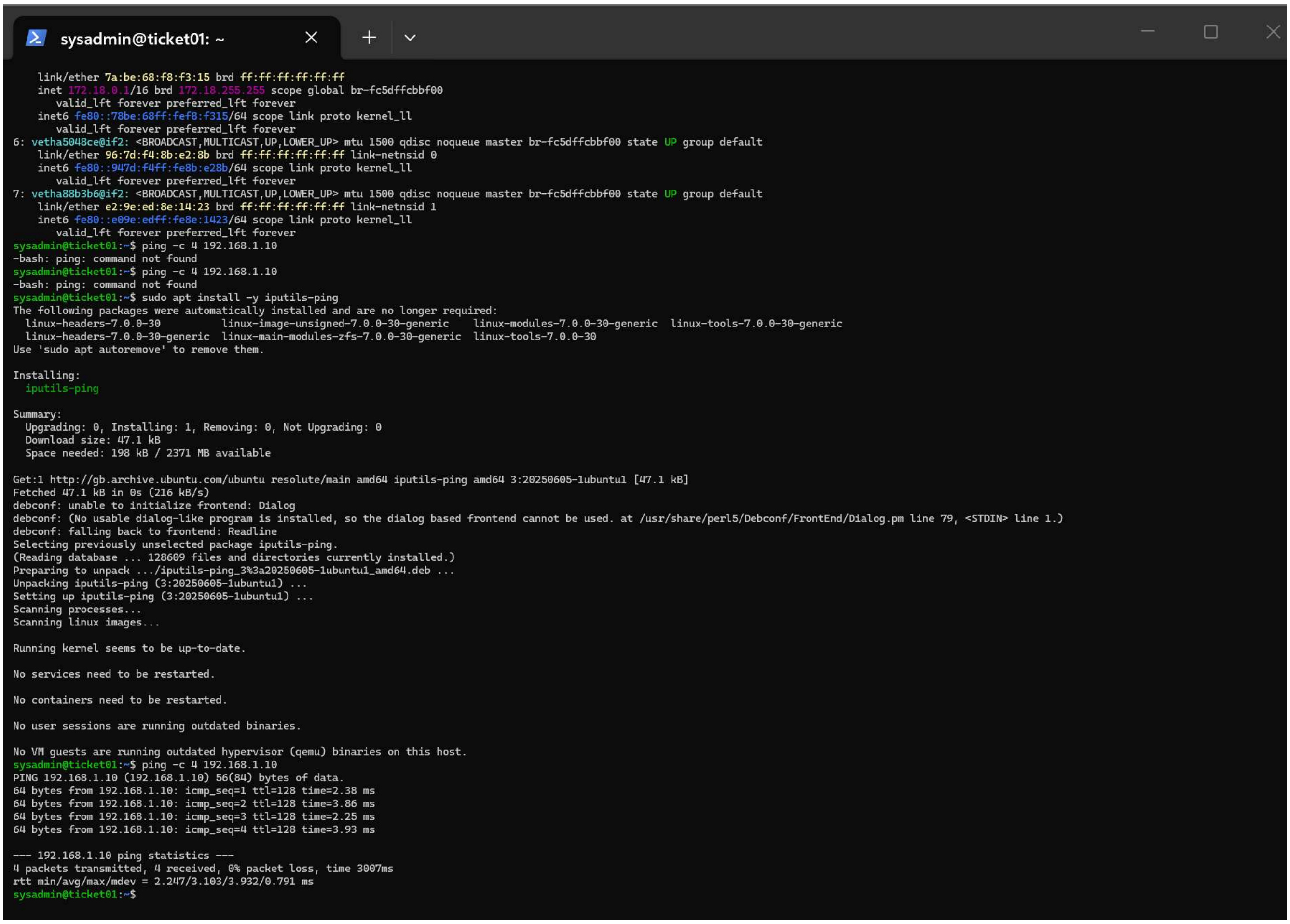


5. Dual-homed networking for LabNetwork integration

A second network adapter (`enp0s8`) was added to TICKET01, attached to the same isolated `LabNetwork` used by DC01 and CLIENT01, and given a static IP (`192.168.1.30/24`) via netplan — continuing the addressing scheme already established in the AD lab (DC01 `.10` , CLIENT01 `.20`). The original NAT adapter (`enp0s3`) was left untouched, so remote management access was never lost.

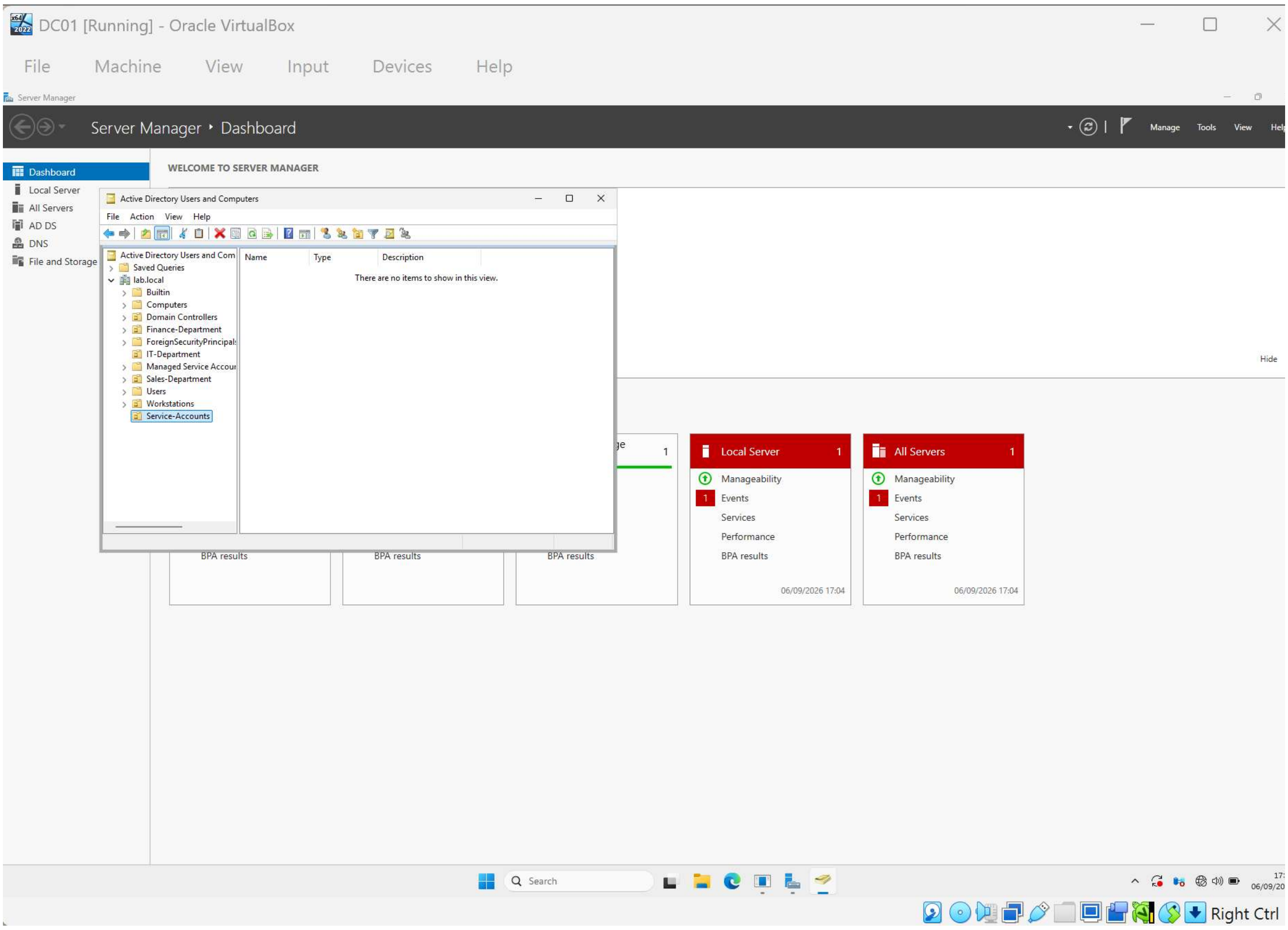

sysadmin@ticket01: ~
inet6 fe80::e8cf:b5ff:fe23:7af0/64 scope link proto kernel_ll
valid_lft forever preferred_lft forever
sysadmin@ticket01:~$ sudo shutdown now
[sudo: authenticate] Password:
sysadmin@ticket01:~$ Connection to 127.0.0.1 closed by remote host.
Connection to 127.0.0.1 closed.
PS C:\Users\jackb> ssh -p 2222 sysadmin@127.0.0.1
ssh: connect to host 127.0.0.1 port 2222: Connection refused
PS C:\Users\jackb> ssh -p 2222 sysadmin@127.0.0.1
sysadmin@127.0.0.1's password:
Welcome to Ubuntu 26.04.1 LTS (GNU/Linux 7.0.0-31-generic x86_64)
* Documentation: https://docs.ubuntu.com
* Management: https://landscape.canonical.com
* Support: https://ubuntu.com/pro
This system has been minimized by removing packages and content that are
not required on a system that users do not log into.
To restore this content, you can run the 'unminimize' command.
Last login: Sun Sep 6 15:25:22 2026 from 10.0.2.2
sysadmin@ticket01:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
inet 127.0.0.1/8 scope host lo
valid_lft forever preferred_lft forever
inet6 ::1/128 scope host noprefixroute
valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
link/ether 08:00:27:76:8f:d2 brd ff:ff:ff:ff:ff:ff
inet 10.0.2.15/24 metric 100 brd 10.0.2.255 scope global dynamic enp0s3
valid_lft 86328sec preferred_lft 86328sec
inet6 fd17:625c:f037:2:a00:27ff:fe76:8fd2/64 scope global dynamic mngtmpaddr noprefixroute
valid_lft 86330sec preferred_lft 14330sec
inet6 fe80::a00:27ff:fe76:8fd2/64 scope link proto kernel_ll
valid_lft forever preferred_lft forever
3: enp0s8: <BROADCAST,MULTICAST> mtu 1500 qdisc noop state DOWN group default qlen 1000
link/ether 08:00:27:08:a1:a5 brd ff:ff:ff:ff:ff:ff
altname enx08002708a1a5
4: docker0: <NO-CARRIER,BROADCAST,MULTICAST,UP> mtu 1500 qdisc noqueue state DOWN group default
link/ether 32:37:38:d8:fb:3d brd ff:ff:ff:ff:ff:ff
inet 172.17.0.1/16 brd 172.17.255.255 scope global docker0
valid_lft forever preferred_lft forever
5: br-fc5dffcbbf00: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue state UP group default
link/ether 7a:be:68:f8:f3:15 brd ff:ff:ff:ff:ff:ff
inet 172.18.0.1/16 brd 172.18.255.255 scope global br-fc5dffcbbf00
valid_lft forever preferred_lft forever
inet6 fe80::78be:68ff:fef8:f315/64 scope link proto kernel_ll
valid_lft forever preferred_lft forever
6: vetha5048ce@if2: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue master br-fc5dffcbbf00 state UP group default
link/ether 96:7d:f4:8b:e2:8b brd ff:ff:ff:ff:ff:ff link-netnsid 0
inet6 fe80::947d:f4ff:fe8b:e28b/64 scope link proto kernel_ll
valid_lft forever preferred_lft forever
7: vetha88b3b6@if2: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue master br-fc5dffcbbf00 state UP group default
link/ether e2:9e:ed:8e:14:23 brd ff:ff:ff:ff:ff:ff link-netnsid 1
inet6 fe80::e09e:edff:fe8e:1423/64 scope link proto kernel_ll
valid_lft forever preferred_lft forever
sysadmin@ticket01:~$

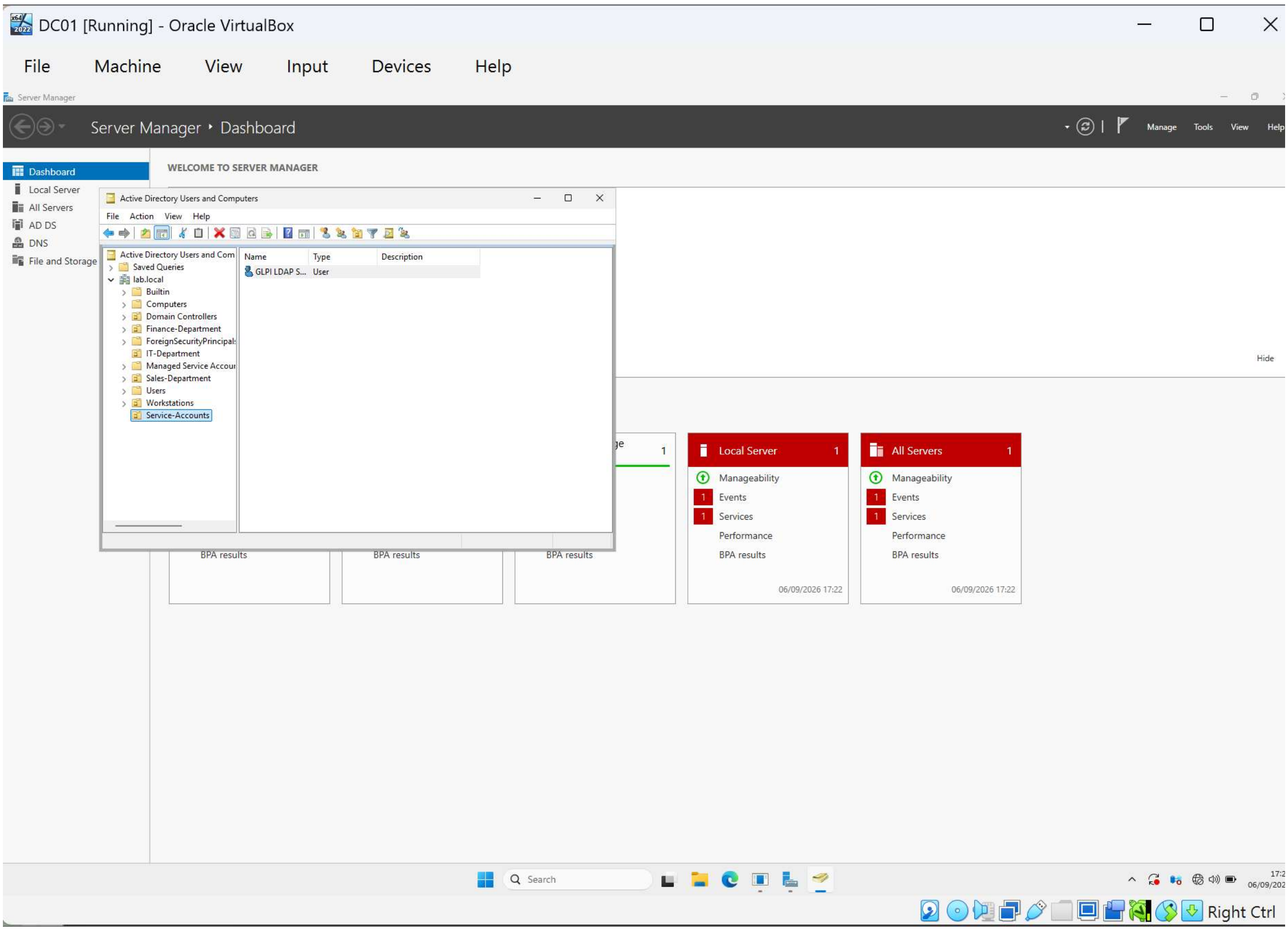

sysadmin@ticket01: ~
sysadmin@ticket01:~$ sudo nano /etc/netplan/50-cloud-init.yaml
[sudo: authenticate] Password:
sysadmin@ticket01:~$ sudo netplan apply
** (configure:1959): WARNING **: 15:54:53.351: Permissions for /etc/netplan/50-cloud-init.yaml are too open. Netplan configuration should NOT be accessible by others.
** (process:1957): WARNING **: 15:54:55.709: Permissions for /etc/netplan/50-cloud-init.yaml are too open. Netplan configuration should NOT be accessible by others.
** (process:1957): WARNING **: 15:54:57.385: Permissions for /etc/netplan/50-cloud-init.yaml are too open. Netplan configuration should NOT be accessible by others.
sysadmin@ticket01:~$ sudo chmod 600 /etc/netplan/50-cloud-init.yaml
sysadmin@ticket01:~$ cat -A /etc/netplan/50-cloud-init.yaml
cat: /etc/netplan/50-cloud-init.yaml: Permission denied
sysadmin@ticket01:~$ sudo cat -A /etc/netplan/50-cloud-init.yaml
network:$
version: 2$
ethernets:$
enp0s3:$
dhcp4: true$
enp0s8:$
dhcp4: no$
addresses: [192.168.1.30/24]$
nameservers:$
addresses: [192.168.1.10]$
sysadmin@ticket01:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
inet 127.0.0.1/8 scope host lo
valid_lft forever preferred_lft forever
inet6 ::1/128 scope host noprefixroute
valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
link/ether 08:00:27:76:8f:d2 brd ff:ff:ff:ff:ff:ff
inet 10.0.2.15/24 metric 100 brd 10.0.2.255 scope global dynamic enp0s3
valid_lft 86156sec preferred_lft 86156sec
inet6 fd17:625c:f037:2:a00:27ff:fe76:8fd2/64 scope global dynamic mngtmpaddr noprefixroute
valid_lft 86328sec preferred_lft 14328sec
inet6 fe80::a00:27ff:fe76:8fd2/64 scope link proto kernel_ll
valid_lft forever preferred_lft forever
3: enp0s8: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
link/ether 08:00:27:08:a1:a5 brd ff:ff:ff:ff:ff:ff
altname enx08002708a1a5
inet 192.168.1.30/24 brd 192.168.1.255 scope global enp0s8
valid_lft forever preferred_lft forever
inet6 fe80::a00:27ff:fe08:a1a5/64 scope link proto kernel_ll
valid_lft forever preferred_lft forever
4: docker0: <NO-CARRIER,BROADCAST,MULTICAST,UP> mtu 1500 qdisc noqueue state DOWN group default
link/ether 32:37:38:d8:fb:3d brd ff:ff:ff:ff:ff:ff
inet 172.17.0.1/16 brd 172.17.255.255 scope global docker0
valid_lft forever preferred_lft forever
5: br-fc5dffcbbf00: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue state UP group default
link/ether 7a:be:68:f8:f3:15 brd ff:ff:ff:ff:ff:ff
inet 172.18.0.1/16 brd 172.18.255.255 scope global br-fc5dffcbbf00
valid_lft forever preferred_lft forever
inet6 fe80::78be:68ff:fef8:f315/64 scope link proto kernel_ll
valid_lft forever preferred_lft forever
6: vetha5048ce@if2: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue master br-fc5dffcbbf00 state UP group default
link/ether 96:7d:f4:8b:e2:8b brd ff:ff:ff:ff:ff:ff link-netnsid 0
inet6 fe80::947d:f4ff:fe8b:e28b/64 scope link proto kernel_ll
valid_lft forever preferred_lft forever
7: vetha88b3b6@if2: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue master br-fc5dffcbbf00 state UP group default
link/ether e2:9e:ed:8e:14:23 brd ff:ff:ff:ff:ff:ff link-netnsid 1
inet6 fe80::e09e:edff:fe8e:1423/64 scope link proto kernel_ll
valid_lft forever preferred_lft forever
sysadmin@ticket01:~$

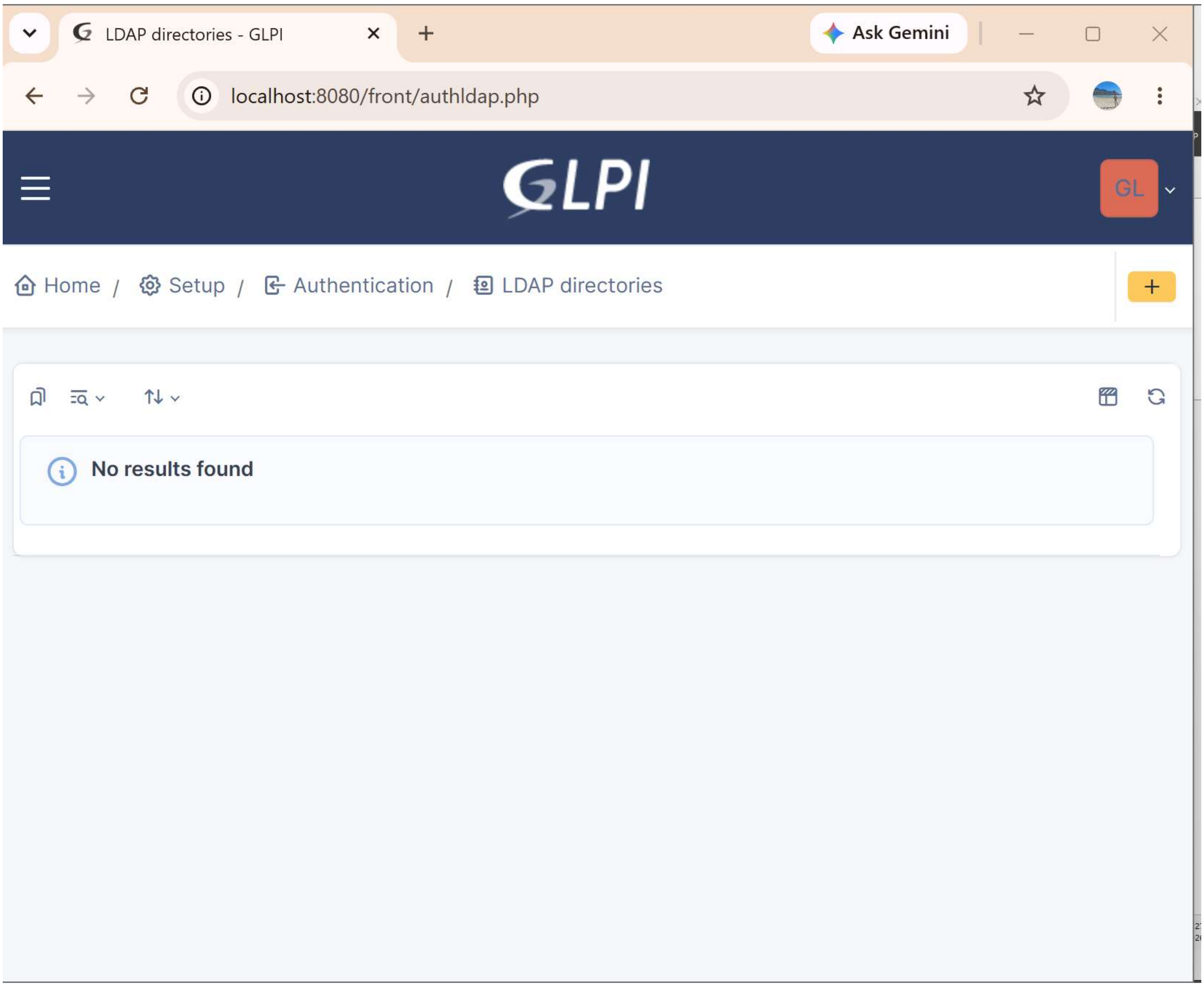


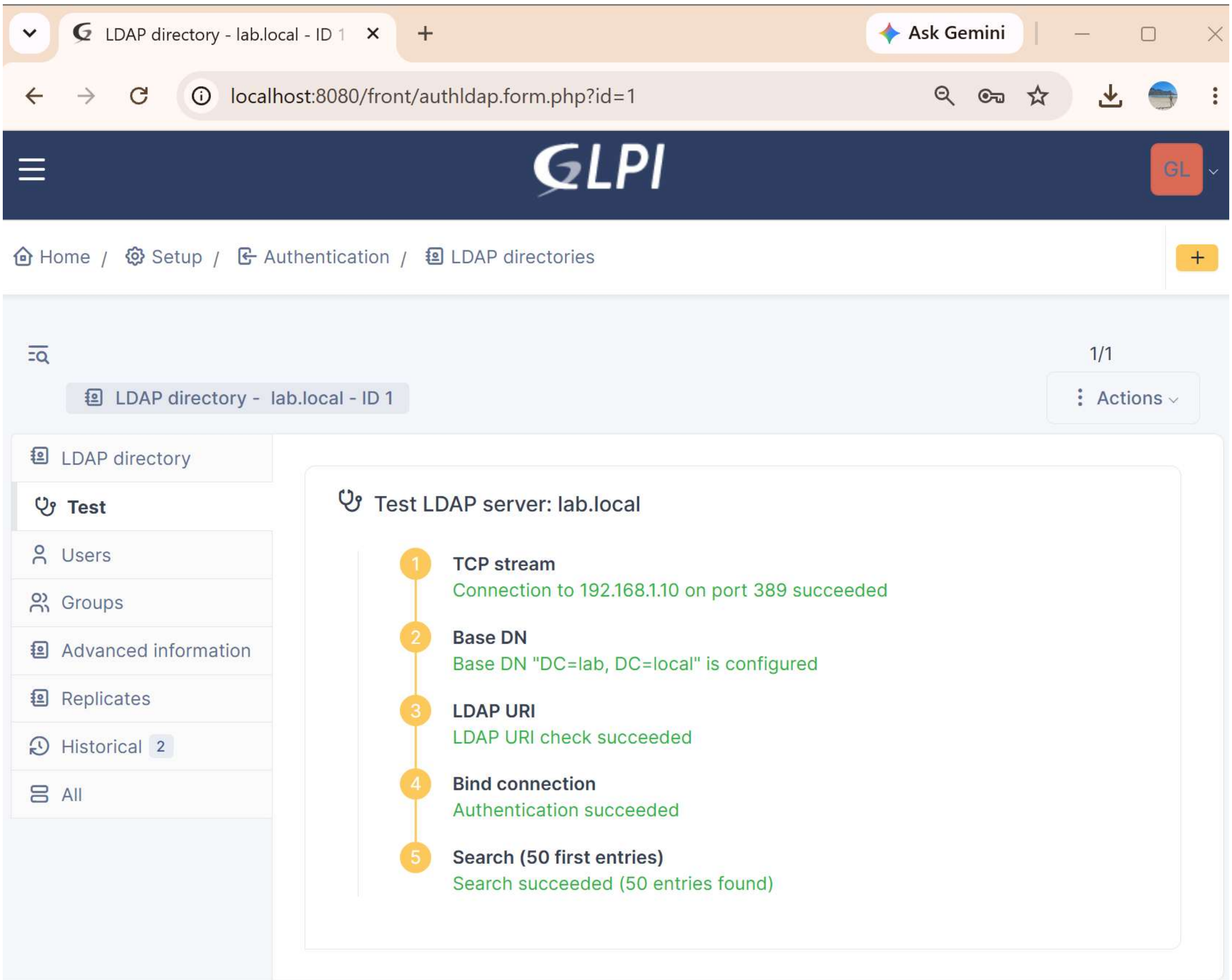
6. LDAP integration with lab.local

A dedicated least-privilege service account (`svc-glpi-ldap`) was created in a new `Service-Accounts` OU on DC01 specifically for this integration, rather than reusing an existing personal account — limiting what an attacker could do if the integration itself were ever compromised. GLPI's LDAP directory was configured to bind to DC01 using this account. The connection initially failed due to DC01's default LDAP signing enforcement (see Troubleshooting Log, Issue 6); once resolved, all five test stages passed, confirming genuine authentication against the domain.

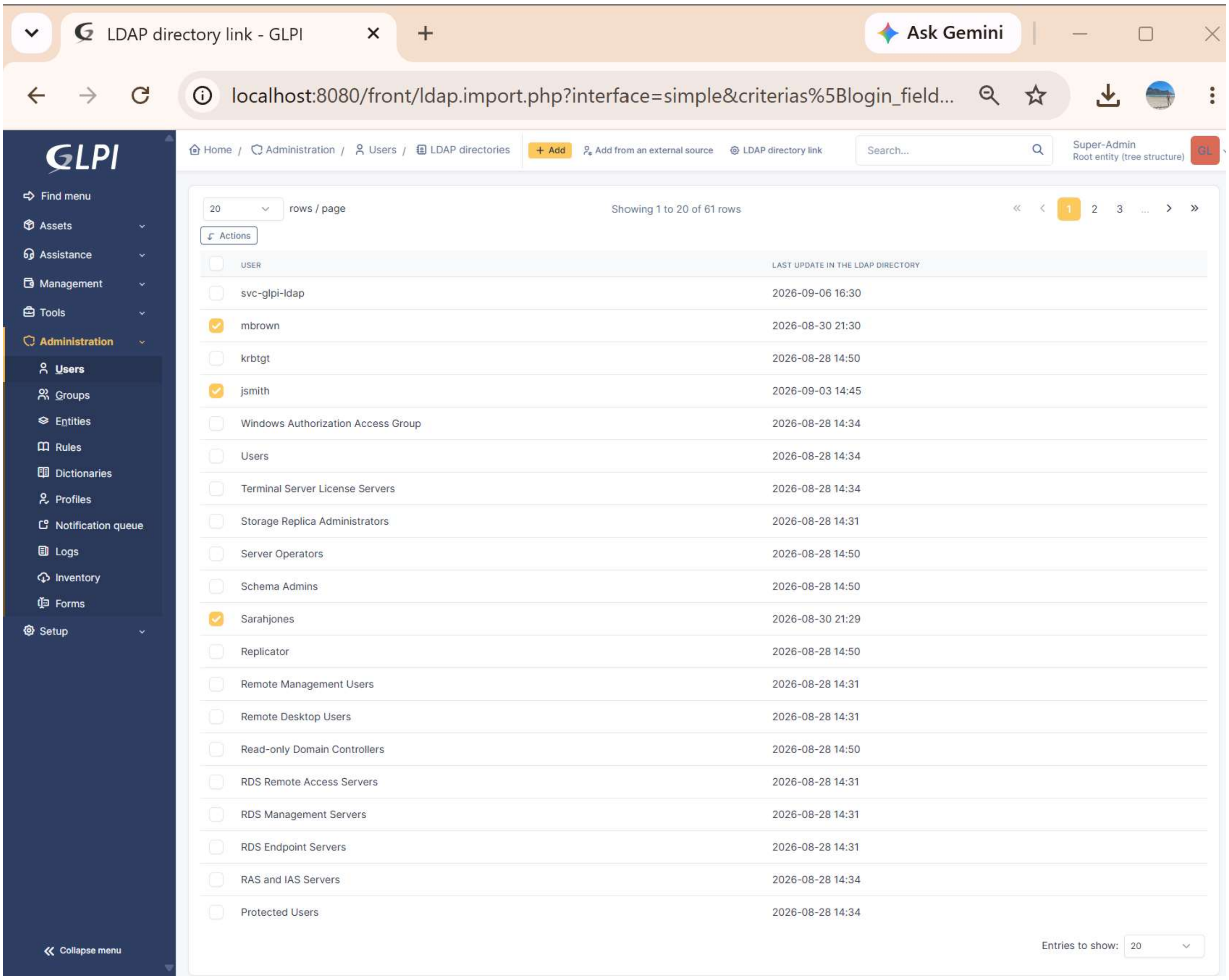


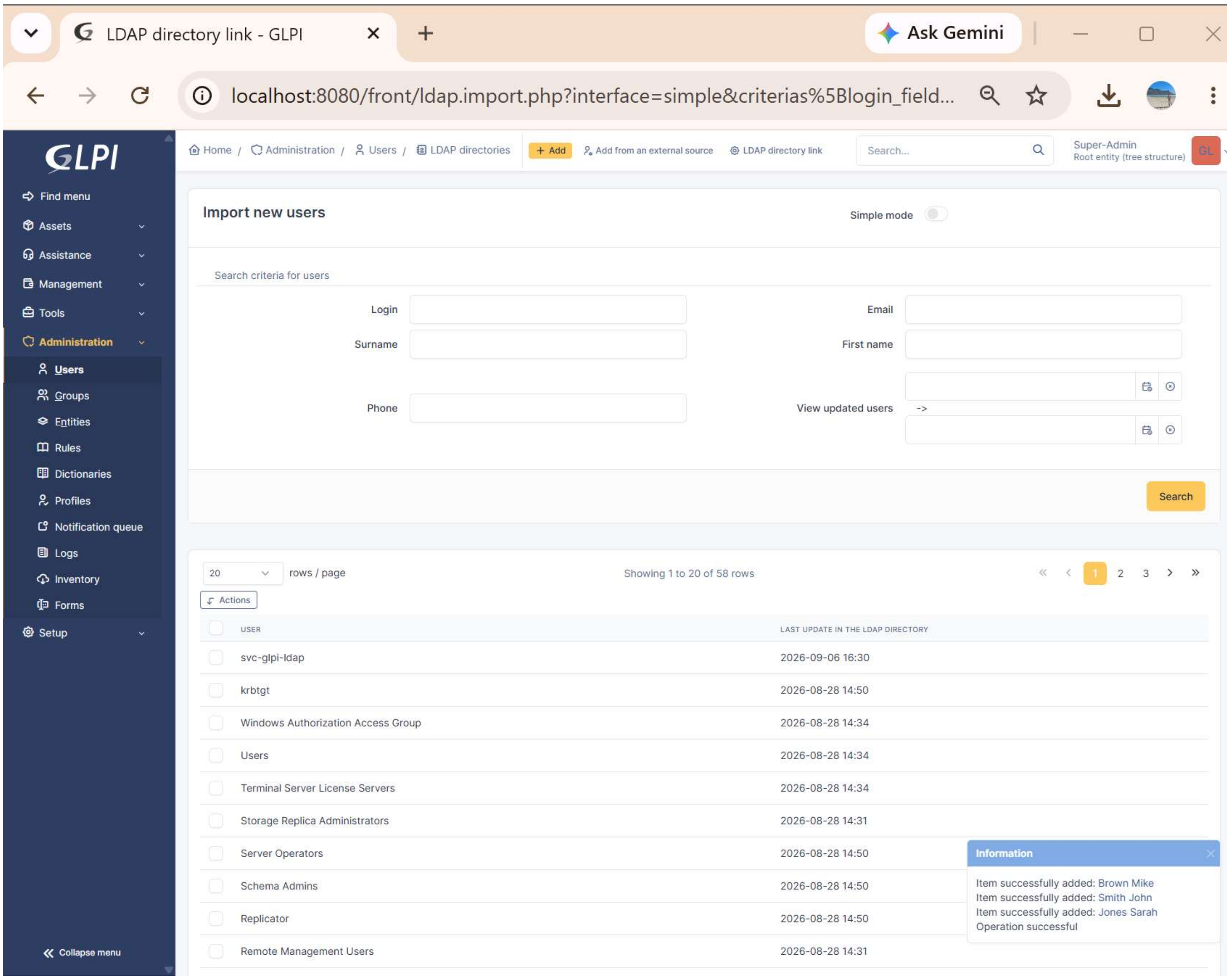


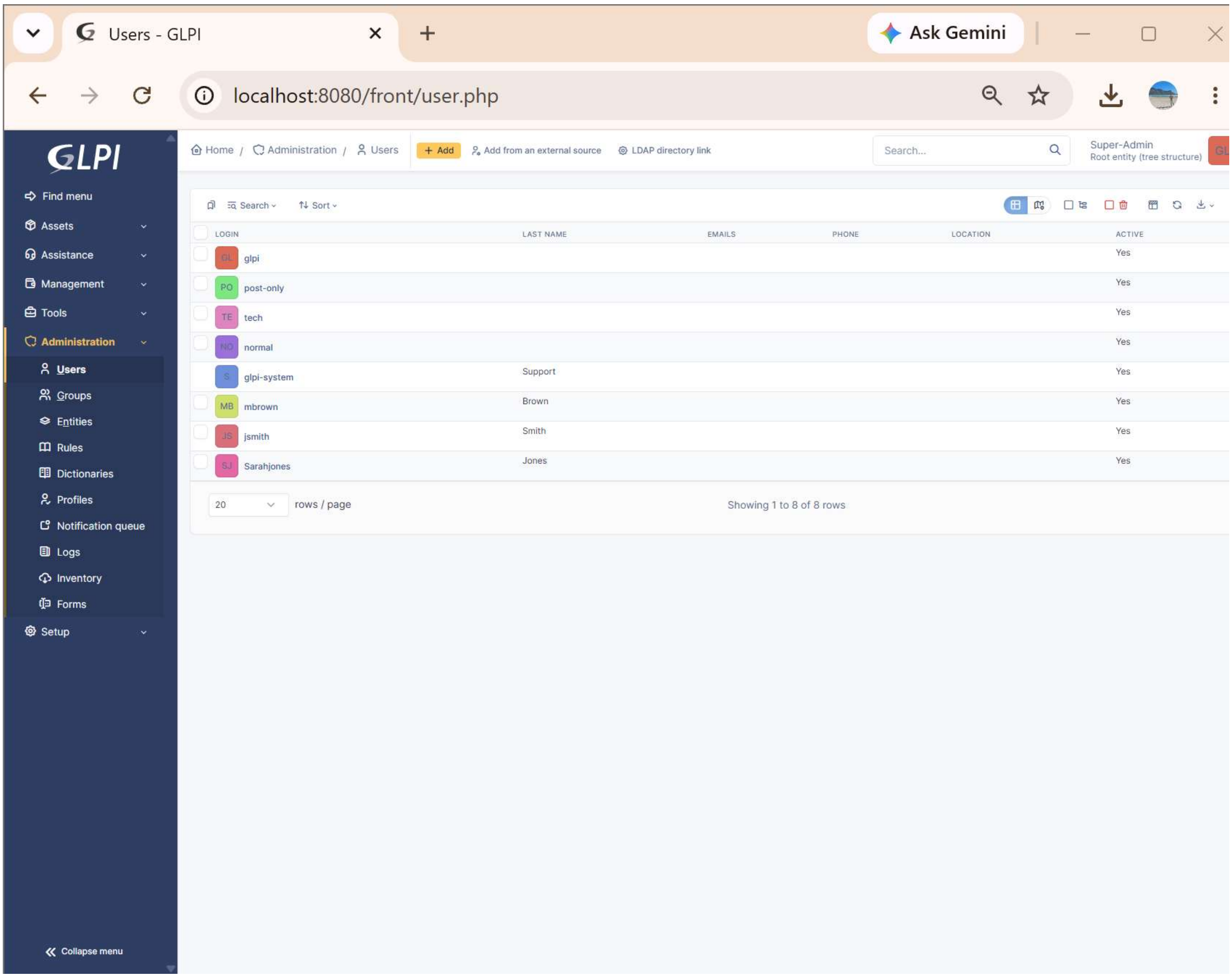

LDAP directories - GLPI
Ask Gemini
localhost:8080/front/authldap.php
GLPI
GL
Home / Setup / Authentication / LDAP directories
No results found



Next: importing jsmith, Sarah Jones, and Mike Brown from their respective OUs as real GLPI users.

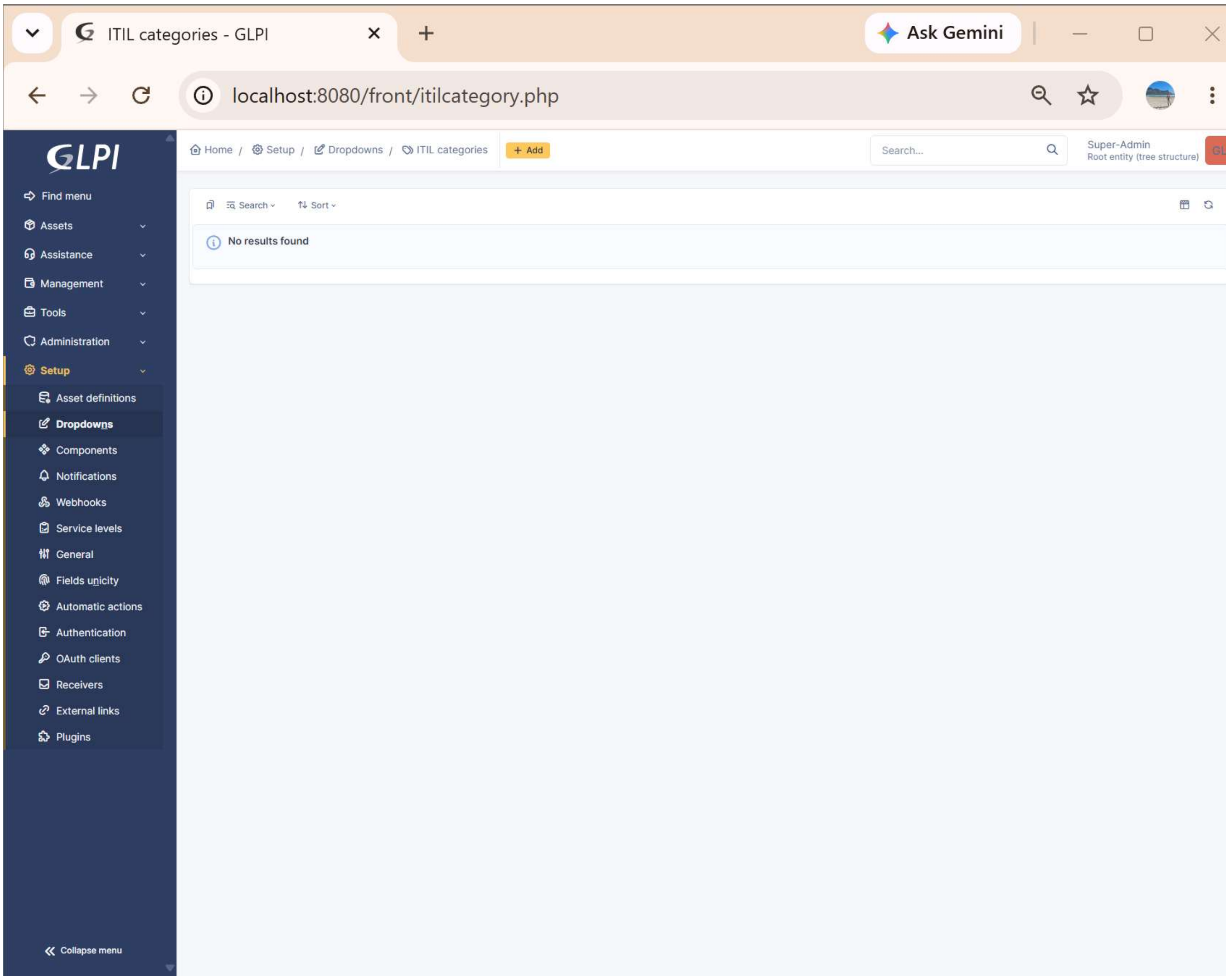

LDAP directory link - GLPI
Ask Gemini
localhost:8080/front/ldap.import.php?interface=simple&criterias%5Blogin_field...
GLPI
Find menu
Assets
Assistance
Management
Tools
Administration
Users
Groups
Entities
Rules
Dictionaries
Profiles
Notification queue
Logs
Inventory
Forms
Setup
Collapse menu
Home / Administration / Users / LDAP directories
+ Add
Add from an external source
LDAP directory link
Search...
Super-Admin
Root entity (tree structure)
20
rows / page
Showing 1 to 20 of 61 rows
1 2 3 ...
Actions
USER
LAST UPDATE IN THE LDAP DIRECTORY
svc-glpi-ldap 2026-09-06 16:30
mbrown 2026-08-30 21:30
krbtgt 2026-08-28 14:50
jsmith 2026-09-03 14:45
Windows Authorization Access Group 2026-08-28 14:34
Users 2026-08-28 14:34
Terminal Server License Servers 2026-08-28 14:34
Storage Replica Administrators 2026-08-28 14:31
Server Operators 2026-08-28 14:50
Schema Admins 2026-08-28 14:50
Sarahjones 2026-08-30 21:29
Replicator 2026-08-28 14:50
Remote Management Users 2026-08-28 14:31
Remote Desktop Users 2026-08-28 14:31
Read-only Domain Controllers 2026-08-28 14:50
RDS Remote Access Servers 2026-08-28 14:31
RDS Management Servers 2026-08-28 14:31
RDS Endpoint Servers 2026-08-28 14:31
RAS and IAS Servers 2026-08-28 14:34
Protected Users 2026-08-28 14:34
Entries to show: 20

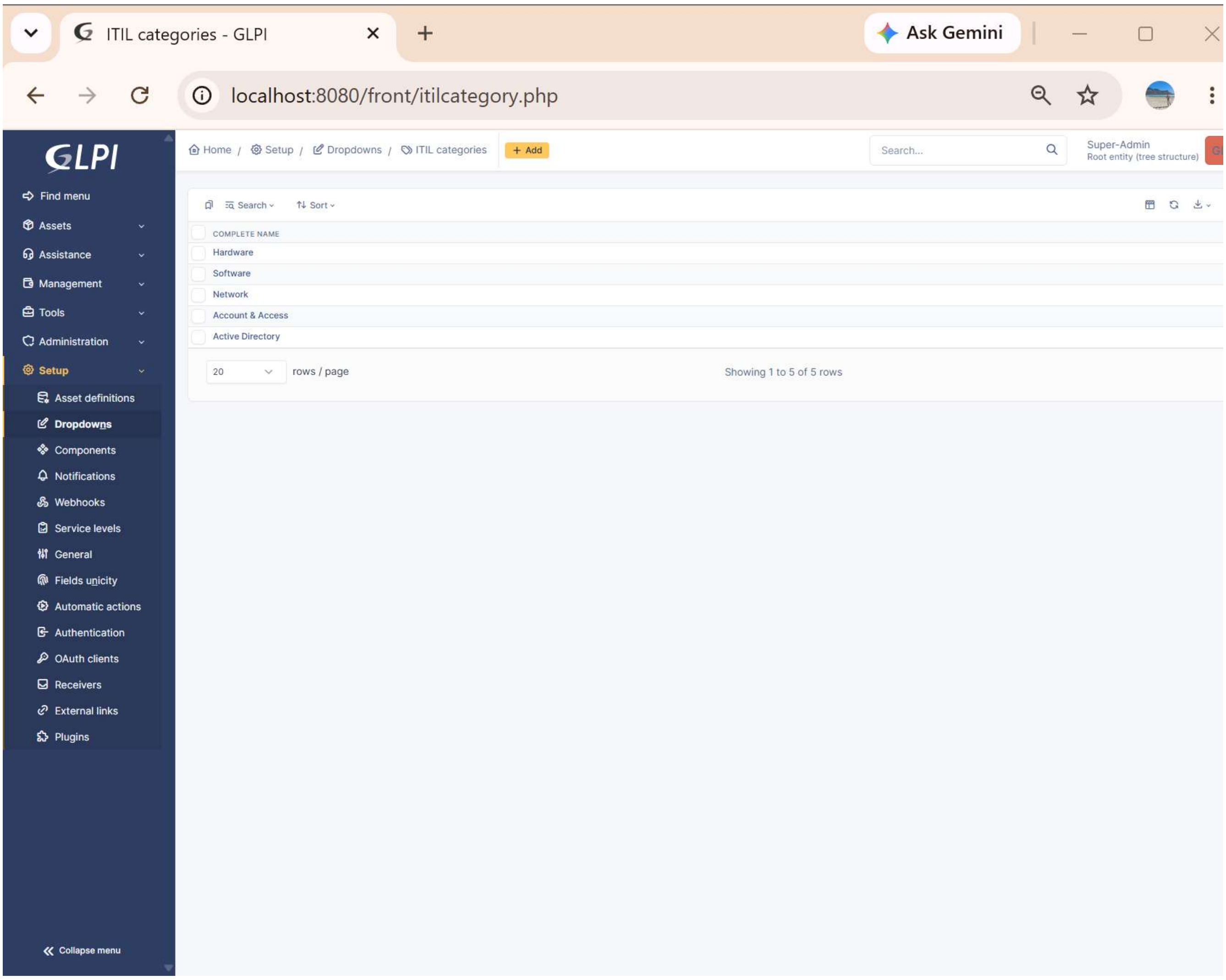

LDAP directory link - GLPI
Ask Gemini
localhost:8080/front/ldap.import.php?interface=simple&criterias%5Blogin_field...
GLPI
Find menu
Assets
Assistance
Management
Tools
Administration
Users
Groups
Entities
Rules
Dictionaries
Profiles
Notification queue
Logs
Inventory
Forms
Setup
Collapse menu
Home / Administration / Users / LDAP directories
+ Add
Add from an external source
LDAP directory link
Search...
Super-Admin
Root entity (tree structure)
Import new users
Simple mode
Search criteria for users
Login
Email
Surname
First name
Phone
View updated users
->
Search
20
rows / page
Showing 1 to 20 of 58 rows
1 2 3
Actions
USER
LAST UPDATE IN THE LDAP DIRECTORY
svc-glpi-ldap 2026-09-06 16:30
krbtgt 2026-08-28 14:50
Windows Authorization Access Group 2026-08-28 14:34
Users 2026-08-28 14:34
Terminal Server License Servers 2026-08-28 14:34
Storage Replica Administrators 2026-08-28 14:31
Server Operators 2026-08-28 14:50
Schema Admins 2026-08-28 14:50
Replicator 2026-08-28 14:50
Remote Management Users 2026-08-28 14:31
Information
Item successfully added: Brown Mike
Item successfully added: Smith John
Item successfully added: Jones Sarah
Operation successful



7. Ticket categories & templates

Five categories created matching real IT support taxonomy, giving tickets a consistent way to be sorted and reported on: Hardware, Software, Network, Account & Access, and Active Directory (the last one specifically for domain/infrastructure-related tickets, tying back to the original AD lab's troubleshooting log).

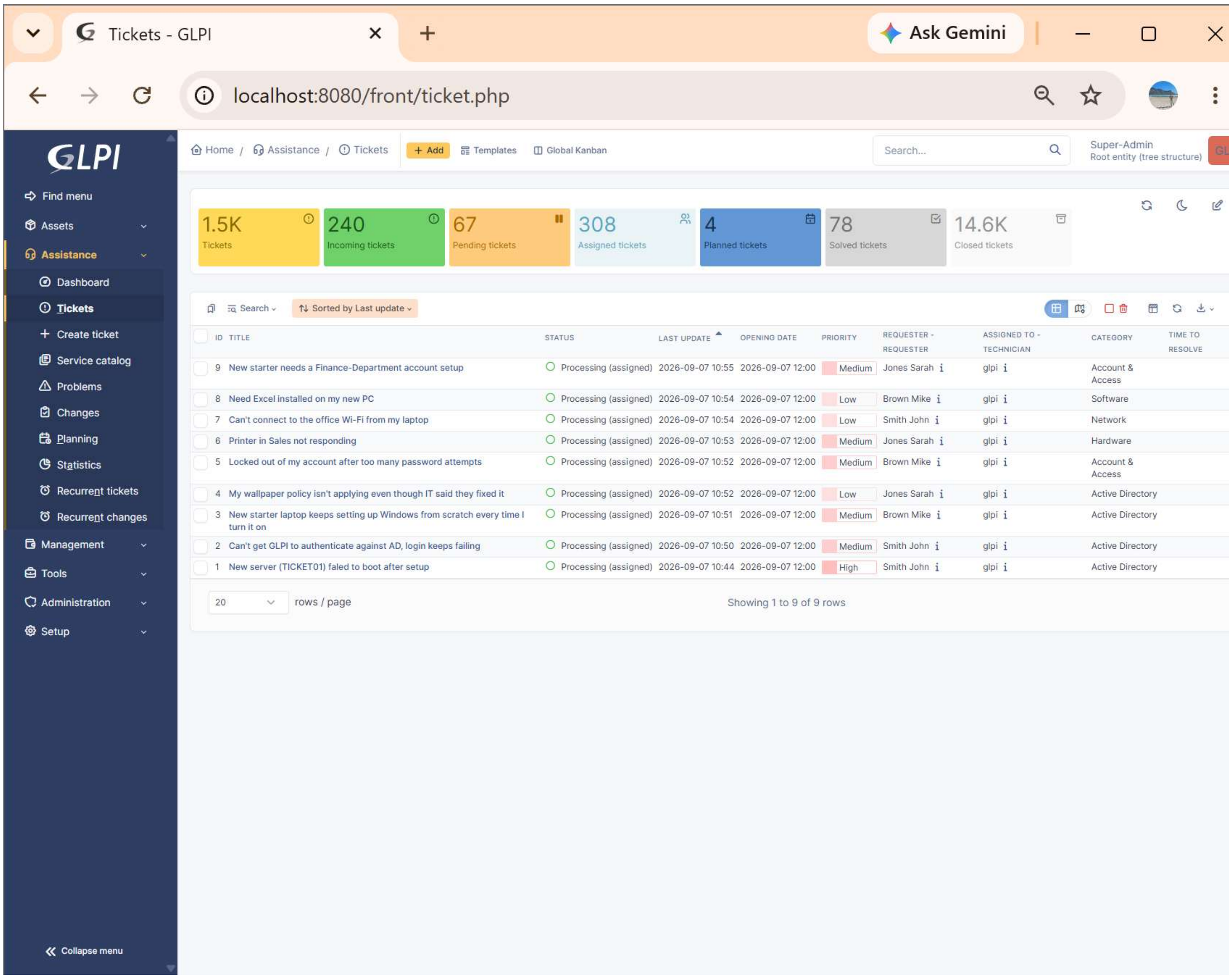

ITIL categories - GLPI
Ask Gemini
localhost:8080/front/itilcategory.php
GLPI
Home / Setup / Dropdowns / ITIL categories
+ Add
Search...
Super-Admin
Root entity (tree structure)
Find menu
Assets
Assistance
Management
Tools
Administration
Setup
Asset definitions
Dropdowns
Components
Notifications
Webhooks
Service levels
General
Fields unicity
Automatic actions
Authentication
OAuth clients
Receivers
External links
Plugins
Collapse menu
Search
Sort
No results found

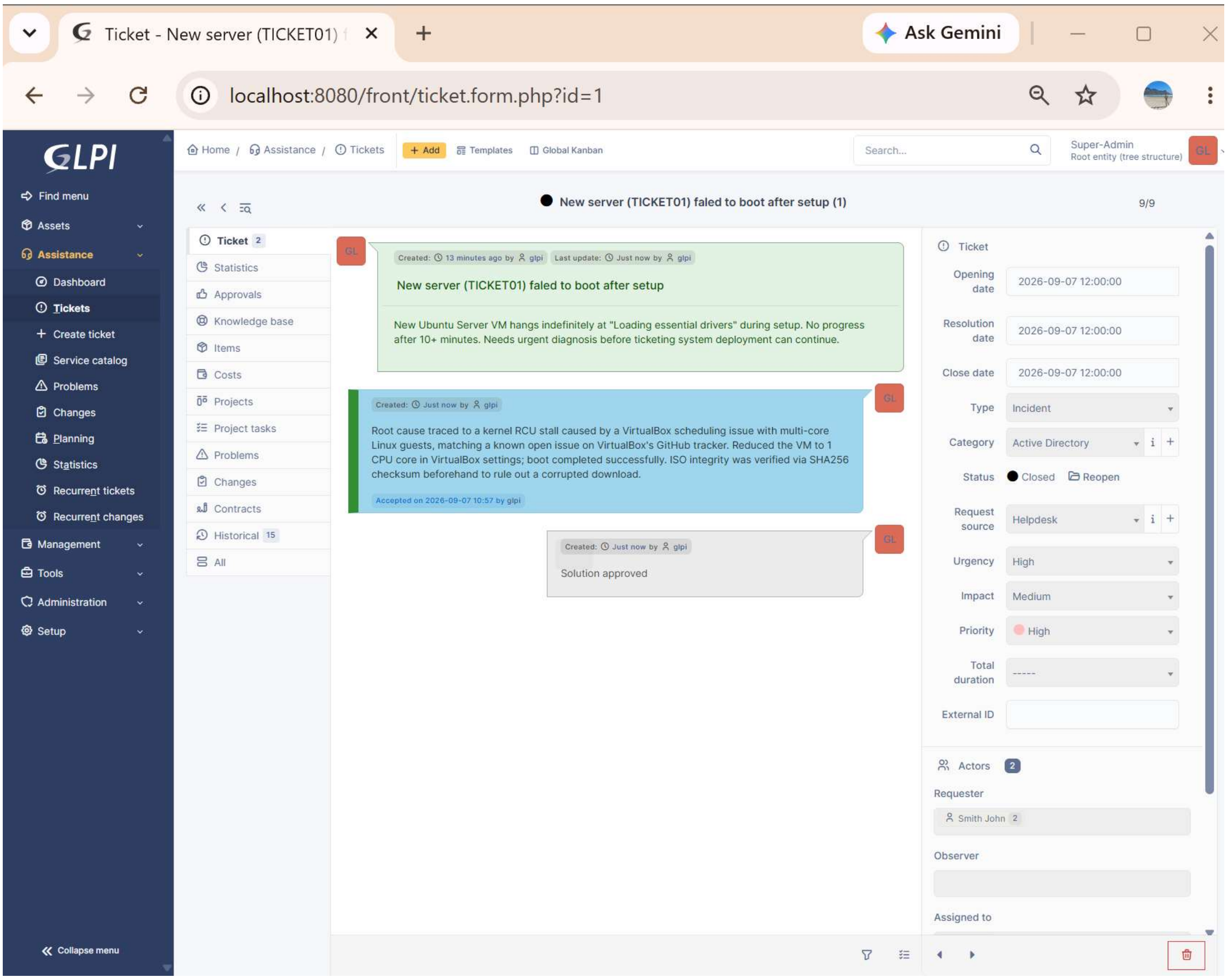


8. Realistic ticket population

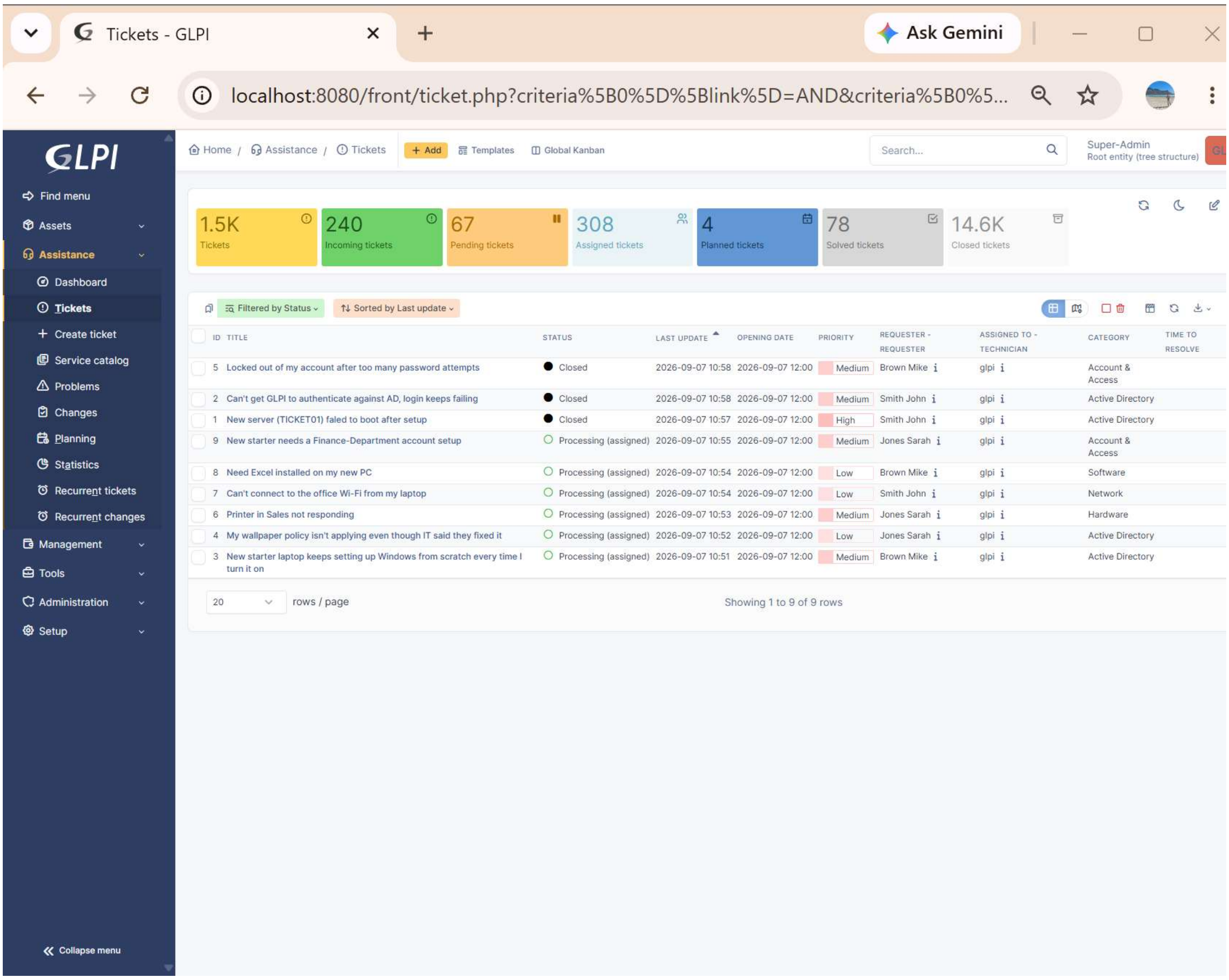
Nine tickets created across all three imported AD users and all five categories, mixing genuine technical incidents drawn from this project's ow troubleshooting log with ordinary day-to-day helpdesk volume. Three tickets (the RCU stall, the LDAP signing failure, and an account lockout) were resolved with real, specific solution text and closed — the remainder left in "Processing (assigned)" status, reflecting a realistic snapshot an active service desk rather than a fully-wrapped demo.

#	Title	Requester	Category	Status
1	New server (TICKET01) failed to boot after setup	jsmith	Active Directory	Closed
2	Can't get GLPI to authenticate against AD, login keeps failing	jsmith	Active Directory	Closed
3	New starter laptop keeps setting up Windows from scratch every time I turn it on	mbrown	Active Directory	Processing
4	My wallpaper policy isn't applying even though IT said they fixed it	Sarahjones	Active Directory	Processing
5	Locked out of my account after too many password attempts	mbrown	Account & Access	Closed
6	Printer in Sales not responding	Sarahjones	Hardware	Processing
7	Can't connect to the office Wi-Fi from my laptop	jsmith	Network	Processing
8	Need Excel installed on my new PC	mbrown	Software	Processing
9	New starter needs a Finance-Department account setup	Sarahjones	Account & Access	Processing


Tickets - GLPI
Ask Gemini
localhost:8080/front/ticket.php
GLPI
Find menu
Assets
Assistance
Dashboard
Tickets
Create ticket
Service catalog
Problems
Changes
Planning
Statistics
Recurrent tickets
Recurrent changes
Management
Tools
Administration
Setup
Collapse menu
Home / Assistance / Tickets
+ Add
Templates
Global Kanban
Search...
Super-Admin
Root entity (tree structure)
1.5K Tickets
240 Incoming tickets
67 Pending tickets
308 Assigned tickets
4 Planned tickets
78 Solved tickets
14.6K Closed tickets
Search
Sorted by Last update
ID TITLE STATUS LAST UPDATE OPENING DATE PRIORITY REQUESTER - REQUESTER ASSIGNED TO - TECHNICIAN CATEGORY TIME TO RESOLVE
9 New starter needs a Finance-Department account setup Processing (assigned) 2026-09-07 10:55 2026-09-07 12:00 Medium Jones Sarah glpi Account & Access
8 Need Excel installed on my new PC Processing (assigned) 2026-09-07 10:54 2026-09-07 12:00 Low Brown Mike glpi Software
7 Can't connect to the office Wi-Fi from my laptop Processing (assigned) 2026-09-07 10:54 2026-09-07 12:00 Low Smith John glpi Network
6 Printer in Sales not responding Processing (assigned) 2026-09-07 10:53 2026-09-07 12:00 Medium Jones Sarah glpi Hardware
5 Locked out of my account after too many password attempts Processing (assigned) 2026-09-07 10:52 2026-09-07 12:00 Medium Brown Mike glpi Account & Access
4 My wallpaper policy isn't applying even though IT said they fixed it Processing (assigned) 2026-09-07 10:52 2026-09-07 12:00 Low Jones Sarah glpi Active Directory
3 New starter laptop keeps setting up Windows from scratch every time I turn it on Processing (assigned) 2026-09-07 10:51 2026-09-07 12:00 Medium Brown Mike glpi Active Directory
2 Can't get GLPI to authenticate against AD, login keeps failing Processing (assigned) 2026-09-07 10:50 2026-09-07 12:00 Medium Smith John glpi Active Directory
1 New server (TICKET01) failed to boot after setup Processing (assigned) 2026-09-07 10:44 2026-09-07 12:00 High Smith John glpi Active Directory
20 rows / page
Showing 1 to 9 of 9 rows

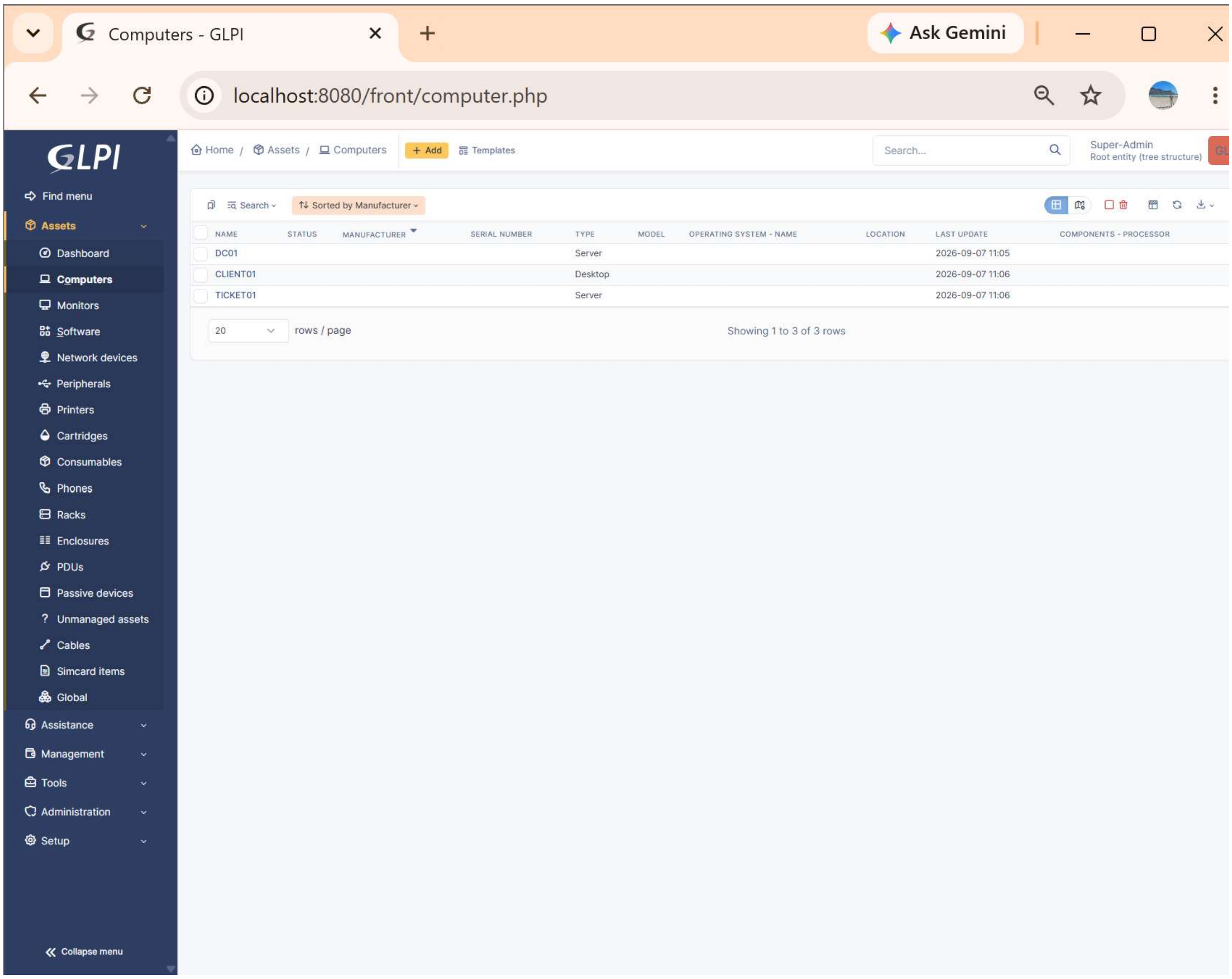

Ticket - New server (TICKET01)
Ask Gemini
localhost:8080/front/ticket.form.php?id=1
GLPI
Find menu
Assets
Assistance
Dashboard
Tickets
Create ticket
Service catalog
Problems
Changes
Planning
Statistics
Recurrent tickets
Recurrent changes
Management
Tools
Administration
Setup
Collapse menu
Home / Assistance / Tickets
+ Add
Templates
Global Kanban
Search...
Super-Admin
Root entity (tree structure)
New server (TICKET01) faled to boot after setup (1)
9/9
Ticket 2
Statistics
Approvals
Knowledge base
Items
Costs
Projects
Project tasks
Problems
Changes
Contracts
Historical 15
All
Created: 13 minutes ago by glpi Last update: Just now by glpi
New server (TICKET01) faled to boot after setup
New Ubuntu Server VM hangs indefinitely at "Loading essential drivers" during setup. No progress after 10+ minutes. Needs urgent diagnosis before ticketing system deployment can continue.
Created: Just now by glpi
Root cause traced to a kernel RCU stall caused by a VirtualBox scheduling issue with multi-core Linux guests, matching a known open issue on VirtualBox's GitHub tracker. Reduced the VM to 1 CPU core in VirtualBox settings; boot completed successfully. ISO integrity was verified via SHA256 checksum beforehand to rule out a corrupted download.
Accepted on 2026-09-07 10:57 by glpi
Created: Just now by glpi
Solution approved
Ticket
Opening date 2026-09-07 12:00:00
Resolution date 2026-09-07 12:00:00
Close date 2026-09-07 12:00:00
Type Incident
Category Active Directory
Status Closed Reopen
Request source Helpdesk
Urgency High
Impact Medium
Priority High
Total duration -----
External ID
Actors 2
Requester
Smith John 2
Observer
Assigned to

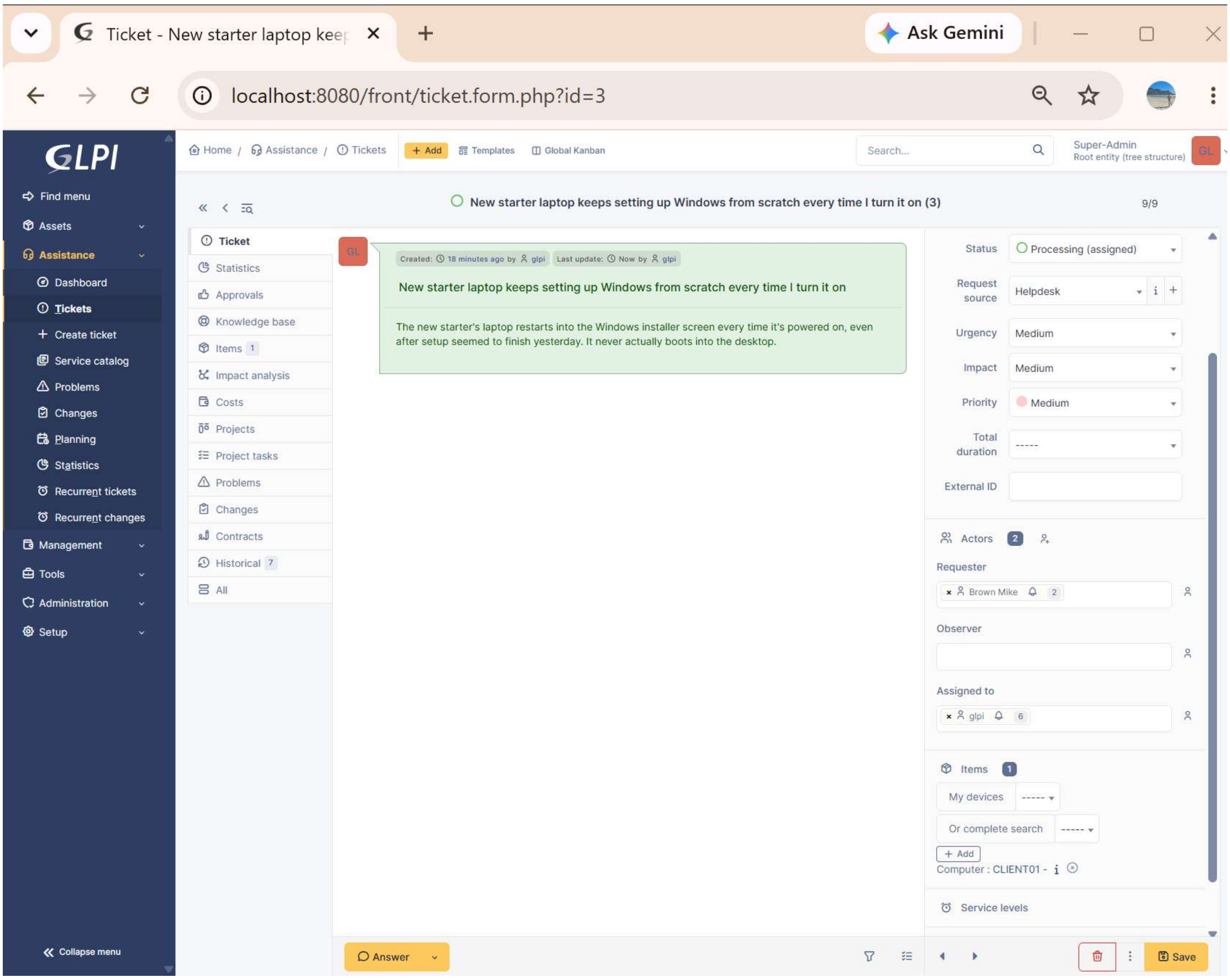
README



9. Asset inventory

GLPI's asset management side was populated with the three actual machines in this lab (DC01, CLIENT01, TICKET01) rather than leaving it empty — since GLPI is an IT asset manager as much as a ticketing tool, an empty inventory would have undersold that half of the product. Ticket #3 was linked directly to its real subject asset (CLIENT01) via the ticket's Items tab, demonstrating GLPI's asset-ticket relationship rather than tickets existing as disconnected text records.


Computers - GLPI
Ask Gemini
localhost:8080/front/computer.php
GLPI
Home / Assets / Computers
+ Add
Templates
Search...
Super-Admin
Root entity (tree structure)
Find menu
Assets
Dashboard
Computers
Monitors
Software
Network devices
Peripherals
Printers
Cartridges
Consumables
Phones
Racks
Enclosures
PDUs
Passive devices
Unmanaged assets
Cables
Simcard items
Global
Assistance
Management
Tools
Administration
Setup
Collapse menu
Search
Sorted by Manufacturer
NAME STATUS MANUFACTURER SERIAL NUMBER TYPE MODEL OPERATING SYSTEM - NAME LOCATION LAST UPDATE COMPONENTS - PROCESSOR
DC01 Server 2026-09-07 11:05
CLIENT01 Desktop 2026-09-07 11:06
TICKET01 Server 2026-09-07 11:06
20 rows / page
Showing 1 to 3 of 3 rows



GLPI's built-in demo data was also disabled, so the dashboard reflects only genuine project data rather than the thousands of placeholder assets and tickets GLPI ships with by default.

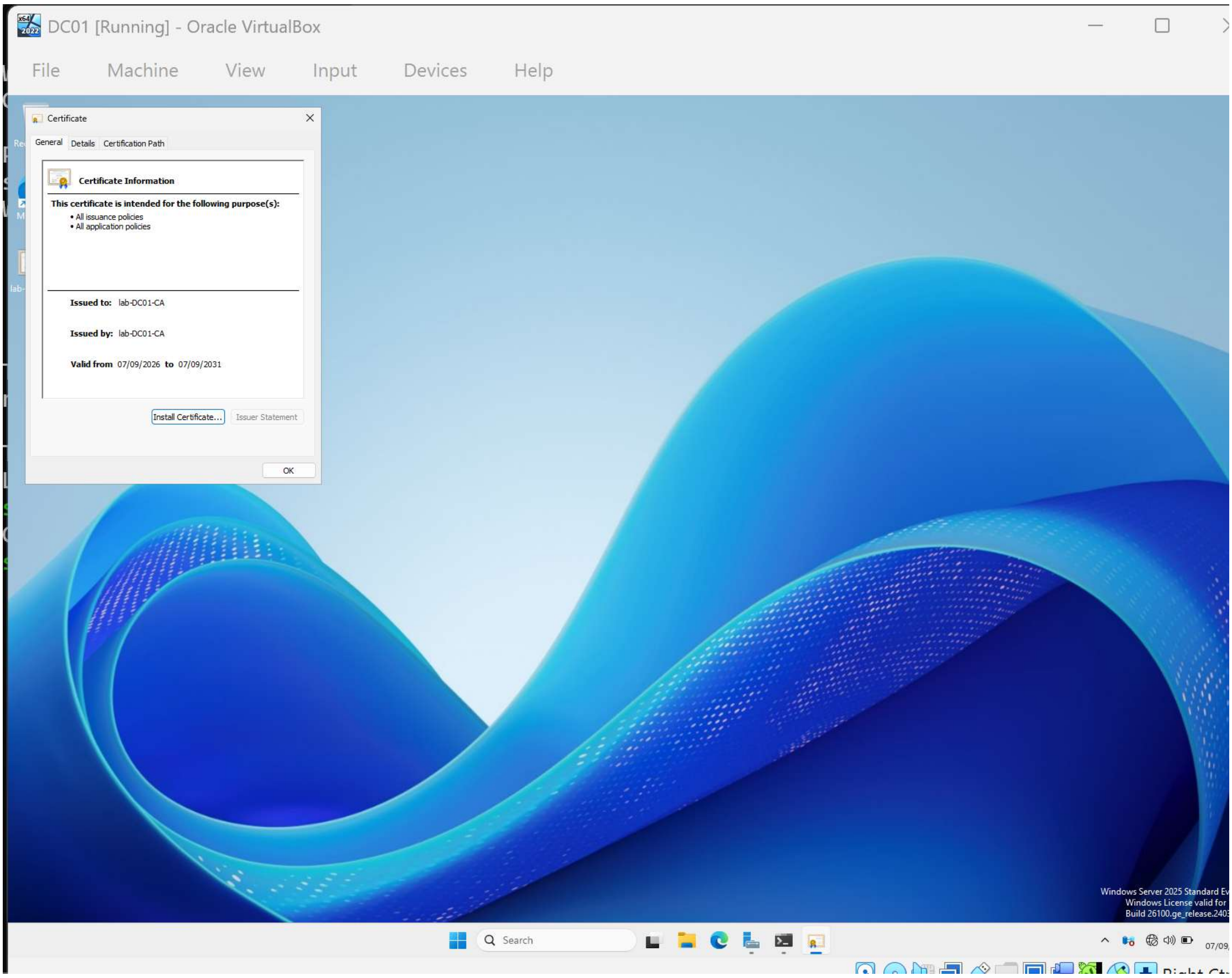
10. LDAPS (stretch goal): replacing the signing workaround with real encryption

The original LDAP integration (Section 6) worked over plain, unencrypted LDAP on port 389, only after relaxing DC01's LDAP signing requirements (see Troubleshooting Log, Issue 6). That was flagged at the time as a deliberate lab simplification rather than a production-corre fix. This stretch goal replaces it with genuine encrypted LDAP (LDAPS, port 636), backed by a real internal Certificate Authority — the fix an actual production environment would require.

Steps completed:

- Installed and configured **Active Directory Certificate Services (AD CS)** on DC01 as an Enterprise Root CA (`lab-DC01-CA`)
- DC01 auto-enrolled itself a valid Domain Controller certificate (`CN=DC01.lab.local`)
- Confirmed port 636 listening on DC01 (`netstat -an | findstr :636`)
- Confirmed TICKET01 could reach it (`nc -zv 192.168.1.10 636` succeeded)
- In GLPI's LDAP directory settings, updated **Port** to `636` and prefixed the **Server** field with `ldaps://` to switch the connection from plain LDAP to LDAPS
- Left **Use TLS** (Advanced information tab) set to **No** — this field is for the separate StartTLS method on port 389, and is mutually exclusive with the `ldaps://` prefix method used here

This surfaced a new, genuinely instructive failure (see Troubleshooting Log, Issue 7), currently being worked through.



Getting the certificate off DC01 required VirtualBox's shared clipboard, set to Bidirectional — the first paste attempt landed directly at the bas prompt instead of inside an open editor (see Troubleshooting Log, Issue 8), corrected by opening `nano` first and pasting into it.

sysadmin@ticket01: ~

```
  GNU nano 8.7.1                        lab-DC01-CA.pem *
```


```
-----BEGIN CERTIFICATE-----
MIIDXzCCAkegAwIBAgIQbSiP3symp7FFru2R6WTmpjANBgkqhkiG9w0BAQsFADBC
MRUwEwYKCZImiZPyLGQBGRYFbG9jYWwxEzARBgoJkiaJk/IsZAEZFgNsYWIxFDAS
BgNVBAMTC2xhYi1EQzAxLUNBMB4XDTI2MDkwNzExMzUxMVoXDTMxMDkwNzExNDUx
MVowQjEVMBMGCgmSJomT8ixkARkWBWxvY2FsMRMwEQYKCZImiZPyLGQBGRYDbGFi
MRQwEgYDVQQDEwtsYWItREMwMS1DQTCCASIwDQYJKoZIhvcNAQEBBQADggEPADCC
AQoCggEBAJiK/RuQmA86YPuHtvUOcEAXv/lqPmXyOLCDi5fcZ4NCME60Rgl47n1h
2rCIj1yq8DN5WfVvV4/IiX2woX2ffV44qW0v3gxXBlAHgxJeTZBIBOdihkyTMN5u
fuZfPDfUrU3gAEMGbGJBZZ8OXhOGhEJqzJpr4cuHkUrFlR7xWgiqdE6AAq1eS365
uMPO4YAHQ6xe18/IupDG1t9ZMPqaSxOuCXi7bJfPJZnvMh08LGwFnnA5e3Bc3AtN
gJsxwh9HvFDfeDtUumUrHEbrkVTJCfxlT+SRaXx9SYU59QwteRTeZ43LfknldpMA
RN2LChOATrlyXf2HhzUfYivajq/rzUECAwEAAaNRME8wCwYDVR0PBAQDAgGGMA8G
A1UdEwEB/wQFMAMBAf8wHQYDVR0OBBYEFNKUbqEpn4+/IEptMmibu2T8rZvrMBAG
CSsGAQQBgjcVAQQDAgEAMA0GCSqGSIb3DQEBCwUAA4IBAQCVJyu257JB31vx+0yY
qMONr53uf0WjA/VbXZapjLiFSgFMeUfbMOyCJ20DRR5U/6Ux8e6PZR6qVgiHPULq
qrvweyuHmkJApPkblXqx1ICM3dhCiq03vjRXBRVS3Ng4tGorXLHbeHpaA3G5u8g1
1/6MplWtcwizcfDWbAELTDeDVtVh42zhvAsXeDeN9x5ztXIXe0djFbD/4iN+Nw9r
YDWYZQGXyYuSjmVSsAZCw+zNgSOSy2r0L6YCGPS44qyKiBdSt+Cl8k9X2EYT2ZRj
Kxvk4TO+YiFDX0MFPngS+4w80LOJ3fKiCAiuESR0rZGK/2Adz1kfdCAWrPXfkP5K
gB9p
-----END CERTIFICATE-----
```


```
^G Help        ^O Write Out   ^F Where Is    ^K Cut         ^T Execute     ^C Location
^X Exit        ^R Read File   ^\ Replace     ^U Paste       ^J Justify     ^/ Go To Line
```

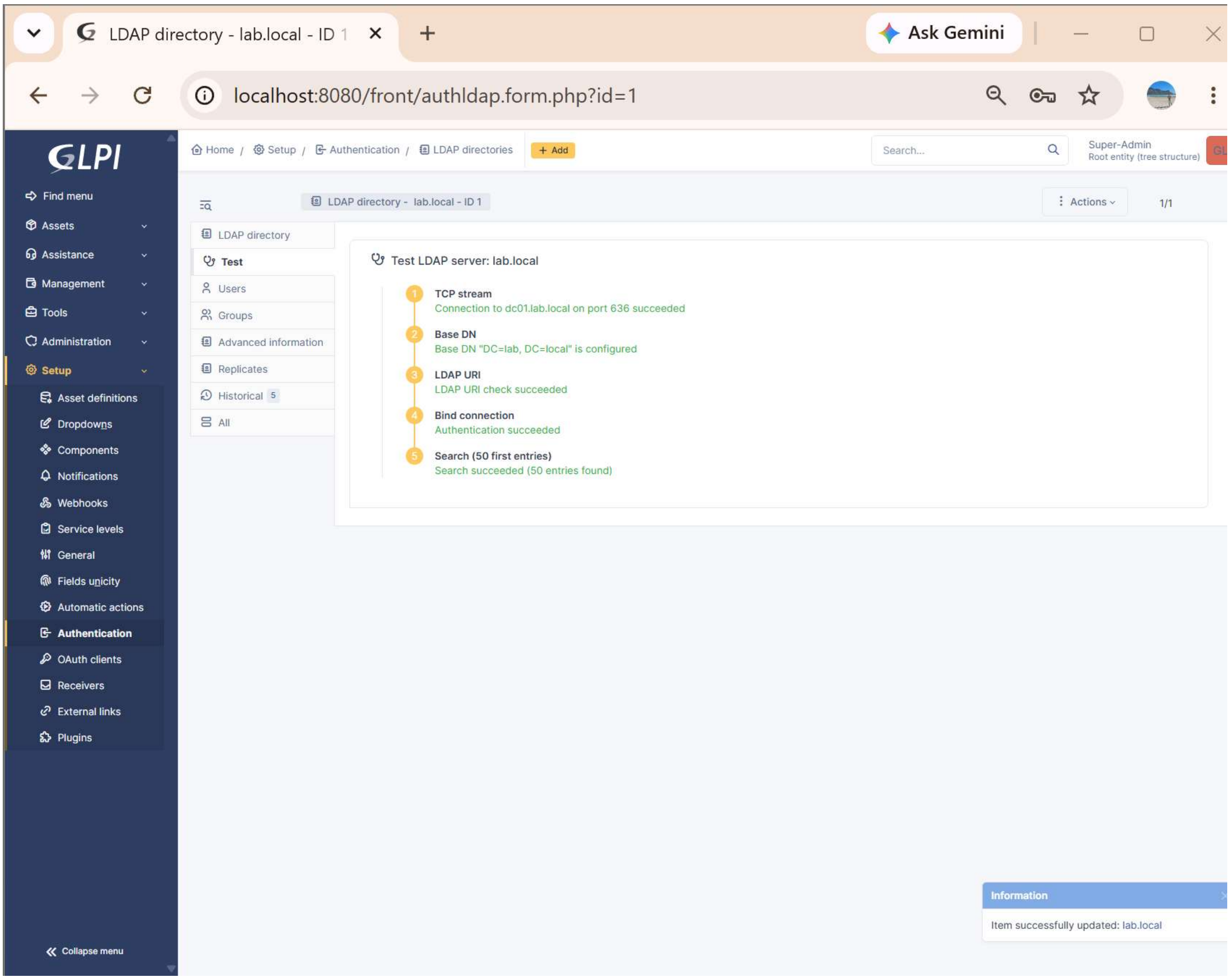
sysadmin@ticket01: ~

```
sysadmin@ticket01:~$    nano lab-DC01-CA.pem
sysadmin@ticket01:~$ cat lab-DC01-CA.pem
-----BEGIN CERTIFICATE-----
MIIDXzCCAkegAwIBAgIQbSiP3symp7FFru2R6WTmpjANBgkqhkiG9w0BAQsFADBC
MRUwEwYKCZImiZPyLGQBGRYFbG9jYWwxEzARBgoJkiaJk/IsZAEZFgNsYWIxFDAS
BgNVBAMTC2xhYi1EQzAxLUNBMB4XDTI2MDkwNzExMzUxMVoXDTMxMDkwNzExNDUx
MVowQjEVMBMGCgmSJomT8ixkARkWBWxvY2FsMRMwEQYKCZImiZPyLGQBGRYDbGFi
MRQwEgYDVQQDEwtsYWItREMwMS1DQTCCASIwDQYJKoZIhvcNAQEBBQADggEPADCC
AQoCggEBAJiK/RuQmA86YPuHtvUOcEAXv/lqPmXyOLCDi5fcZ4NCME60Rgl47n1h
2rCIj1yq8DN5WfVvV4/IiX2woX2ffV44qW0v3gxXBlAHgxJeTZBIBOdihkyTMN5u
fuZfPDfUrU3gAEMGbGJBZZ8OXhOGhEJqzJpr4cuHkUrFlR7xWgiqdE6AAq1eS365
uMPO4YAHQ6xe18/IupDG1t9ZMPqaSxOuCXi7bJfPJZnvMh08LGwFnnA5e3Bc3AtN
gJsxwh9HvFDfeDtUumUrHEbrkVTJCfxlT+SRaXx9SYU59QwteRTeZ43LfknldpMA
RN2LChOATrlyXf2HhzUfYivajq/rzUECAwEAAaNRME8wCwYDVR0PBAQDAgGGMA8G
A1UdEwEB/wQFMAMBAf8wHQYDVR0OBBYEFNKUbqEpn4+/IEptMmibu2T8rZvrMBAG
CSsGAQQBgjcVAQQDAgEAMA0GCSqGSIb3DQEBCwUAA4IBAQCVJyu257JB31vx+0yY
qMONr53uf0WjA/VbXZapjLiFSgFMeUfbMOyCJ20DRR5U/6Ux8e6PZR6qVgiHPULq
qrvweyuHmkJApPkblXqx1ICM3dhCiq03vjRXBRVS3Ng4tGorXLHbeHpaA3G5u8g1
1/6MplWtcwizcfDWbAELTDeDVtVh42zhvAsXeDeN9x5ztXIXe0djFbD/4iN+Nw9r
YDWYZQGXyYuSjmVSsAZCw+zNgSOSy2r0L6YCGPS44qyKiBdSt+Cl8k9X2EYT2ZRj
Kxvk4TO+YiFDX0MFPngS+4w80LOJ3fKiCAiuESR0rZGK/2Adz1kfdCAWrPXfkP5K
gB9p
-----END CERTIFICATE-----
sysadmin@ticket01:~$ docker ps
CONTAINER ID   IMAGE              COMMAND                  CREATED        STATUS
 PORTS                                    NAMES
260bbffc46e1   glpi/glpi:latest   "/opt/glpi/entrypoin…"   22 hours ago   Up 3 hours
 0.0.0.0:80->80/tcp, [::]:80->80/tcp   glpi-glpi-1
c46961918a01   mysql              "docker-entrypoint.s…"   22 hours ago   Up 3 hours
 3306/tcp, 33060/tcp                      glpi-db-1
sysadmin@ticket01:~$
```

sysadmin@ticket01: ~

```
PS C:\WINDOWS\System32> ssh sysadmin@localhost -p 2222
The authenticity of host '[localhost]:2222 ([127.0.0.1]:2222)' can't be established.
ED25519 key fingerprint is SHA256:zSQZY2uuy4EV6ywC9co18PYQIm0bVW+E+B8IA4oEOno.
This host key is known by the following other names/addresses:
    C:\Users\jackb/.ssh/known_hosts:1: [127.0.0.1]:2222
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '[localhost]:2222' (ED25519) to the list of known hosts.
sysadmin@localhost's password:
Permission denied, please try again.
sysadmin@localhost's password:
Welcome to Ubuntu 26.04.1 LTS (GNU/Linux 7.0.0-31-generic x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

This system has been minimized by removing packages and content that are
not required on a system that users do not log into.

To restore this content, you can run the 'unminimize' command.
Last login: Mon Sep  7 09:44:56 2026 from 10.0.2.2
sysadmin@ticket01:~$ docker exec -u root glpi-glpi-1 mkdir -p /etc/ldap
sysadmin@ticket01:~$ docker cp lab-DC01-CA.pem glpi-glpi-1:/etc/ldap/lab-DC01-CA.pem
Successfully copied 3.07kB to glpi-glpi-1:/etc/ldap/lab-DC01-CA.pem
sysadmin@ticket01:~$ docker exec -u root -it glpi-glpi-1 bash
root@260bbffc46e1:/var/www/glpi# echo "TLS_CACERT /etc/ldap/lab-DC01-CA.pem" >> /etc/l
dap/ldap.conf
cat /etc/ldap/ldap.conf
exit
TLS_CACERT /etc/ldap/lab-DC01-CA.pem
exit
sysadmin@ticket01:~$
```

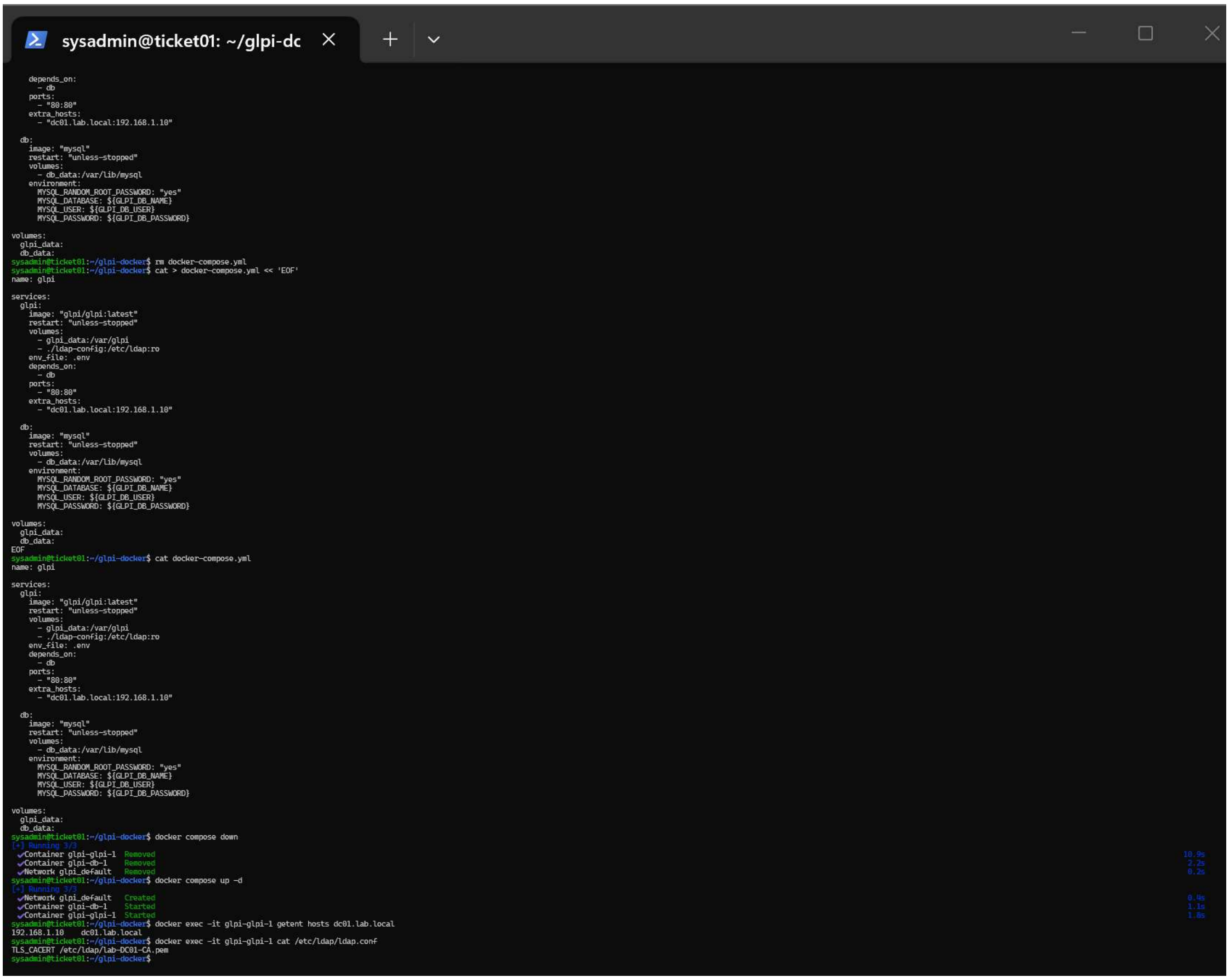
Trusting the CA alone wasn't enough — the container also couldn't resolve DC01 by hostname, and GLPI's Server field was pointing at DC01's IP address rather than the hostname on its certificate, causing a TLS hostname-verification mismatch (see Troubleshooting Log, Issues 9-10). Adding a direct `/etc/hosts` entry inside the container and switching the Server field to `ldaps://dc01.lab.local` resolved it completely.



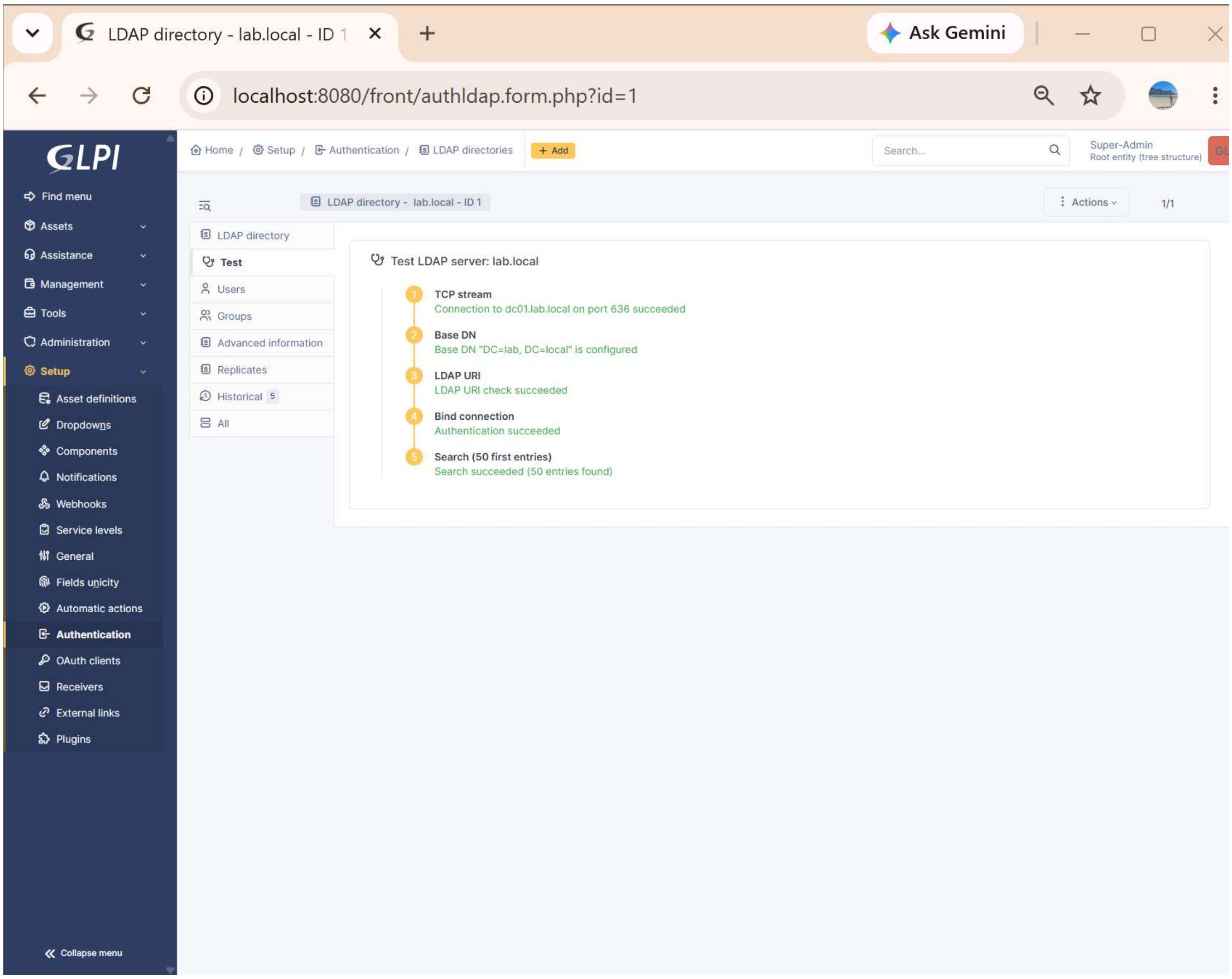
GLPI now authenticates against `lab.local` over genuinely encrypted LDAPS (port 636), backed by a real internal CA — replacing the original plain-LDAP-with-relaxed-signing workaround from Section 6 with the production-correct fix.

Known limitation: the container-side changes (`TLS_CACERT` in `/etc/ldap/ldap.conf` , the `/etc/hosts` entry) were applied live via `docker exec` and are not yet persisted anywhere in `docker-compose.yml` or a mounted volume. They will be lost if the `glpi` container is ever rebuilt from scratch (`docker compose down && up` , not just `restart`). Baking these into the compose file (a config volume mount plus an `extra_hosts` entry) is a natural next step to make the fix durable rather than a one-off runtime patch.

Update — made persistent. The runtime-only fix above was replaced with a proper `docker-compose.yml` configuration: a bind-mounted `./ldap-config` folder (containing `lab-DC01-CA.pem` and a small `ldap.conf` with the `TLS_CACERT` directive) mounted read-only to `/etc/ldap` inside the container, plus an `extra_hosts` entry mapping `dc01.lab.local` to `192.168.1.10` so the container doesn't need to rely on Docker's own DNS resolution. This was verified against a full container rebuild — not just a restart — to prove it genuinely survives recreation rather than just surviving a soft restart:



With that confirmed, the LDAPS stretch goal is complete: GLPI authenticates against `lab.local` over genuinely encrypted LDAP on port 636, backed by a real internal CA (`lab-DC01-CA`), with the entire trust and hostname-resolution configuration defined declaratively in `docker-compose.yml` rather than living only inside a running container's memory. This replaces the original plain-LDAP-with-relaxed-signing-requirements workaround from Section 6 with the actual production-correct fix that workaround was standing in for.



Troubleshooting Log

Real issues encountered and resolved during the build, included deliberately — diagnosing and fixing genuine problems, not just following steps that work first time, is what actually demonstrates capability.

#	Issue	Root Cause	Resolution
1	TICKET01 VM registered in VirtualBox but failed to boot with `VERR_FILE_NOT_FOUND`, "Could not open the medium"	The virtual hard disk (`.vdi`) file never actually finished being created during VM creation, despite the VM's configuration files registering successfully — a silent failure in disk provisioning rather than a visible error at creation time	Removed the broken VM registration (Remove → Delete all files), recreated the VM, and explicitly inspected the "Specify virtual hard disk" panel before clicking Finish to confirm the disk path, size, and type were populated correctly before proceeding
2	TICKET01 boot hung indefinitely at "Loading essential drivers" during Ubuntu 26.04.1 install, with no further progress after 10+ minutes	Kernel-level RCU stall (`rcu: INFO: rcu_preempt detected stalls on CPUs/tasks`) caused by a timing/scheduling desync between VirtualBox 7.2.x and multi-core Linux guests — matched to a known, currently open issue on VirtualBox's public GitHub tracker affecting the same version family	Reduced the VM's allocated CPU count from 2 to 1 (Settings → System → Processor). VM booted cleanly immediately afterward. Before concluding this was the cause, ruled out: a corrupted ISO download (verified via SHA256 checksum against Canonical's official published hash — confirmed exact match), incorrect hardware virtualization/acceleration settings, and host-level CPU/disk resource contention (host CPU utilisation was only 22% at time of hang)
3	`docker compose up -d` failed immediately with `permission denied while trying to connect to`	Docker's daemon is controlled by root and communicates through a restricted Unix socket. A freshly created non-root user is not	Added the user to the `docker` group with `sudo usermod -aG docker $USER`, then fully logged out and reconnected via SSH, since

#	Issue	Root Cause	Resolution
	the Docker daemon socket at `unix:///var/run/docker.sock`	automatically granted access to that socket — membership of the `docker` group has to be explicitly assigned	group membership changes only take effect on a new login session rather than the current one. Confirmed fixed by successfully re-running `docker compose up -d`, which pulled both images and started the `glpi` and `db` containers
4	Browser could not reach GLPI at `http://localhost` (`ERR_CONNECTION_REFUSED`), despite `curl -I http://localhost` succeeding *inside* the VM	The only NAT port-forwarding rule configured was for SSH (port 22). Port 80 (GLPI's web server) had no forwarding rule, so the host's browser had no route in, even though the container itself was working correctly	Added a second NAT port-forwarding rule (host `8080` → guest `80`) in VirtualBox's Network settings. GLPI became reachable at `http://localhost:8080`
5	Second network adapter's static IP (`192.168.1.30`) failed to apply; `netplan apply` errored with `unknown key 'enp0s8'`	While editing the netplan YAML file in `nano`, the existing file content was accidentally overwritten rather than appended to, leaving only the new block with incorrect indentation (not nested under `network:` / `ethernets:`)	Rebuilt `/etc/netplan/50-cloud-init.yaml` from scratch with the full, correctly-nested structure for both `enp0s3` and `enp0s8`. Verified the exact file structure with `cat -A` (showing whitespace/line-endings explicitly) before reapplying, rather than trusting the editor's on-screen display. A follow-on `chmod 600` permissions fix briefly blocked the verifying user's own read access to the file; resolved by reading it with `sudo`
6	GLPI's LDAP connection test failed with `Authentication failed: Strong(er) authentication required(8)` at the Bind connection stage	Windows Server 2025 (DC01's OS) enforces LDAP server signing requirements by default, rejecting unsigned/unencrypted bind attempts over plain LDAP (port 389) — a stricter default than older Windows Server versions used. Network connectivity (`nc -zv 192.168.1.10 389`) and the account credentials were both confirmed correct beforehand, isolating the cause specifically to this security policy	Relaxed "**Domain controller: LDAP server signing requirements**" to `None` and "**...Enforcement**" to `Disabled` via the Default Domain Controllers Policy in Group Policy Management, then forced the change with `gpupdate /force`. This is a deliberate, documented simplification appropriate for an isolated lab without a certificate authority — the production-correct fix would be LDAPS (encrypted LDAP over port 636) with a properly issued certificate, which was considered but deferred as a larger, separate task
7	After switching to LDAPS (`ldaps://192.168.1.10`, port 636), GLPI's connection test passed **TCP stream**, **Base DN**, and **LDAP URI** stages, then failed at **Bind connection** with `Authentication failed: Can't contact LDAP server(-1)`	The raw TCP socket to port 636 opens fine (which is why the earlier stages pass), but the actual TLS handshake is rejected during the bind attempt because the OpenLDAP client library used by PHP/GLPI — running inside the `glpi` Docker container, which has its own isolated filesystem and certificate trust store — has never been told to trust the newly created `lab-DC01-CA` root certificate. The container's default CA bundle only contains public, well-known root CAs, not this lab's private internal CA	Exported `lab-DC01-CA`'s root certificate from DC01 in Base64 (PEM) format, copied it into the `glpi` container's filesystem, and registered it as trusted via `TLS_CACERT` in the container's `/etc/ldap/ldap.conf` (see Issues 8-10 for the full path to a working fix)
8	Pasting the exported certificate text into TICKET01's SSH session landed the raw Base64 content directly at the bash `$` prompt instead of inside a file, after fixing VirtualBox's shared clipboard direction (DC01 → host)	The clipboard paste itself worked correctly, but `nano lab-DC01-CA.pem` had not been opened first — pasting multi-line text straight at an active shell prompt causes bash to interpret each line as a separate command rather than storing it as file content	No damage occurred — each garbled Base64 line simply returned a harmless "command not found," since none matched a real command. Fixed by pressing `Ctrl+C` to clear the prompt, opening the target file with `nano lab-DC01-CA.pem` first, then pasting the still-available clipboard content into the open editor before saving
9	`docker exec -it glpi-glpi-1 mkdir -p /etc/ldap` failed with `Permission denied`; later, running `docker cp` / `docker exec`	`docker exec` connects as the container image's default user, `www-data`, which lacks permission to create directories under `/etc`. Separately, `docker` is a host-level CLI tool —	Exited back to TICKET01's own host shell, then re-ran the container commands with `docker exec -u root glpi-glpi-1 ...` to gain root privileges inside the container

#	Issue	Root Cause	Resolution
	commands from inside an already-open container shell failed with `bash: docker: command not found`	it isn't installed inside the container itself, so it can't be run from a shell that's already inside the container	for the `mkdir` and the `TLS_CACERT` configuration step
10	After successfully adding `TLS_CACERT` inside the container, GLPI's LDAPS test still failed at **Bind connection** with the identical `Authentication failed: Can't contact LDAP server(-1)`	The CA is now trusted, ruling that cause out — the remaining suspect is a hostname/certificate mismatch. GLPI's Server field connects via DC01's IP address (`ldaps://192.168.1.10`), but the certificate's subject is `CN=DC01.lab.local` . TLS hostname verification compares the connection address against the certificate's name, and an IP-vs-hostname mismatch fails that check regardless of CA trust. Confirming this, `docker exec -it glpi-glpi-1 getent hosts dc01.lab.local` returned nothing — the container couldn't resolve the hostname at all, since Docker containers use their own DNS resolution separate from the host VM's netplan/DNS settings	Added a direct entry to the container's `/etc/hosts` (`docker exec -u root glpi-glpi-1 bash -c 'echo "192.168.1.10 dc01.lab.local" >> /etc/hosts'`) so the hostname would resolve, then changed GLPI's Server field to `ldaps://dc01.lab.local` to match the certificate's subject name. All five test stages passed on retest, including a full directory search
11	While rewriting `docker-compose.yml` to persist the LDAPS fix, a `cat > file << 'EOF'` heredoc paste produced a corrupted file with the YAML duplicated and a garbled line in the middle	The heredoc's closing `EOF` was pasted with no line break before a second, duplicate copy of the same command — merging into one line (`EOFcat > docker-compose.yml << 'EOF'`). Since a heredoc terminator must appear alone on its own line, bash didn't recognize this as the end of input, so it kept treating everything afterward (the garbled line, then a second full copy of the YAML) as file content	Deleted the corrupted file (`rm docker-compose.yml`) and re-ran the exact same heredoc as a single, uninterrupted paste, confirming the result with `cat` before proceeding — one clean copy of the YAML, no duplication

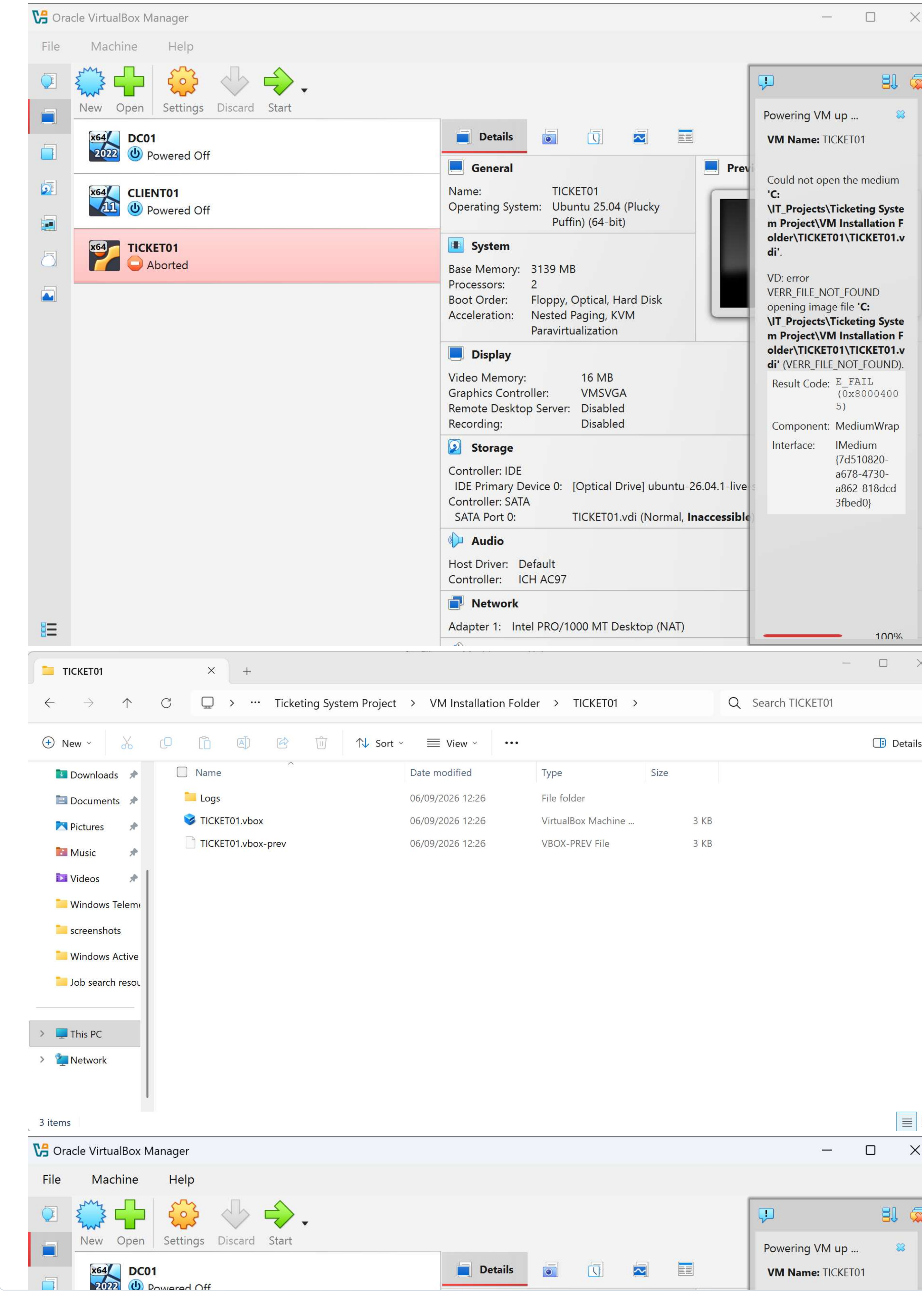
Evidence

Screenshots are split into two numbered sequences for clarity:

- `doc-NN` — standard build steps, working as expected
- `ts-NN` — errors, warnings, or troubleshooting steps and their fixes

Full-resolution screenshots are in the `screenshots` folder; the images above and below are embedded directly from there.

Issue 1: Missing virtual disk file



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